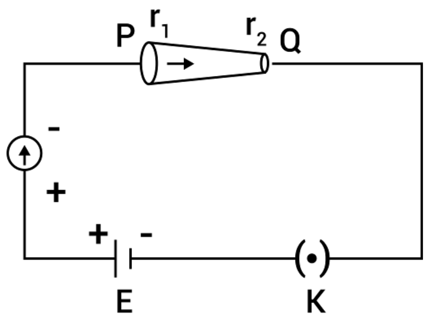
Q1: In the given figure, a battery of emf E is connected across a conductor PQ of length 'l' and different area of cross – sections having radii r_1 and r_2 $(r_2 < r_1)$.



Choose the correct option as one moves from P to Q:

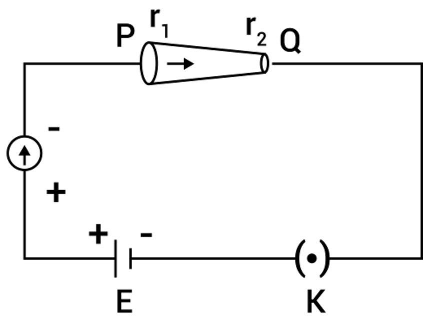
(A) Drift velocity of electron increases.

(B) Electric field decreases.

(C) Electron current decreases.

(D) All of these

Q1: In the given figure, a battery of emf E is connected across a conductor PQ of length 'l' and different area of cross – sections having radii $r_1 \; and \; r_2 \; (r_2 < r_1)$.



Choose the correct option as one moves from P to Q:

(A) Drift velocity of electron increases.

(B) Electric field decreases.

(C) Electron current decreases.

(D) All of these

Solution:

On moving from P to Q
Current density increase
$\mathrm{J} = \sigma \mathrm{E}$
Electric field increase
Hence, Drift velocity increases.

Q2: The number of molecules in one litre of an ideal gas at 300 K and 2 atmospheric pressure with mean kinetic energy $2 \times 10^{-9} J$ per molecules is:

(A) 0.75×10^{11}

(B) 3×10^{11}

(C) 1.5×10^{11}

(D) 6×10^{11}

Q2: The number of molecules in one litre of an ideal gas at 300 K and 2 atmospheric pressure with mean kinetic energy $2 \times 10^{-9} J$ per molecules is:

(A) 0.75×10^{11}

(B) 3×10^{11}

(C) 1.5×10^{11}

(D) 6×10^{11}

Solution:

$\mathrm{KE} = \frac{3}{2}\mathrm{kT}$

$\mathrm{PV} = \frac{\mathrm{N}}{\mathrm{N_A}}\mathrm{RT}$

$\mathrm{N} = \frac{\mathrm{PV}}{\mathrm{kT}}$

$= \mathrm{N} = 1.5 \times 10^{11}$

Q3: The relative permittivity of distilled water is 81. The velocity of light in it will be: (Given $\mu_r = 1$)

(A) $4.33 \times 10^7 \ m/s$

(B) $2.33 \times 10^7 m/s$

(C) $3.33 \times 10^7 \ m/s$

(D) $5.33 \times 10^7 m/s$

Q3: The relative permittivity of distilled water is 81. The velocity of light in it will be: (Given $\mu_r = 1$)

(A) $4.33 \times 10^7\ m/s$

(B) $2.33 \times 10^7 m/s$

(C) $3.33 \times 10^7\ m/s$

(D) $5.33 \times 10^7 m/s$

Solution:

$$v = \frac{c}{\sqrt{\mu_r \varepsilon_r}} = \frac{3 \times 10^8\ \mathrm{m/s}}{\sqrt{1 \times 81}}$$
$$= \frac{1}{3} \times 10^8\ \mathrm{m/s}$$
$$= 3.33 \times 10^7\ \mathrm{m/s}$$

Q4:

List-I	List-II
(a) MI of the rod (length L, Mass M, about an axis $\perp$ to the rod passing through the midpoint)	(i) $8ML^2/3$
(b) MI of the rod (length L, Mass 2M, about an axis $\perp$ to the rod passing through one of its end)	(ii) $ML^2/3$
(c) MI of the rod (length 2L, Mass M, about an axis $\perp$ to the rod passing through its midpoint)	(iii) $ML^2/12$
(d) MI of the rod (Length 2L, Mass 2M, about an axis $\perp$ to the rod passing through one of its end)	(iv) $2\ ML^2/3$

Choose the correct answer from the options given below:

(A) (a) – (ii), (b) – (iii), (c) – (i), (d) – (iv)

(B) (a) – (ii), (b) – (i), (c) – (iii), (d) – (iv)

(C) (a) – (iii), (b) – (iv), (c) – (ii), (d) – (i)

(D) (a) – (iii), (b) – (iv), (c) – (i), (d) – (ii)

Q4:

List-I	**List-II**
(a) MI of the rod (length L, Mass M, about an axis $\perp$ to the rod passing through the midpoint)	(i) $8ML^2/3$
(b) MI of the rod (length L, Mass 2M, about an axis $\perp$ to the rod passing through one of its end)	(ii) $ML^2/3$
(c) MI of the rod (length 2L, Mass M, about an axis $\perp$ to the rod passing through its midpoint)	(iii) $ML^2/12$
(d) MI of the rod (Length 2L, Mass 2M, about an axis $\perp$ to the rod passing through one of its end)	(iv) $2\ ML^2/3$

Choose the correct answer from the options given below:

(A) (a) – (ii), (b) – (iii), (c) – (i), (d) – (iv)

(B) (a) – (ii), (b) – (i), (c) – (iii), (d) – (iv)

(C) (a) – (iii), (b) – (iv), (c) – (ii), (d) – (i)

(D) (a) – (iii), (b) – (iv), (c) – (i), (d) – (ii)

Solution:

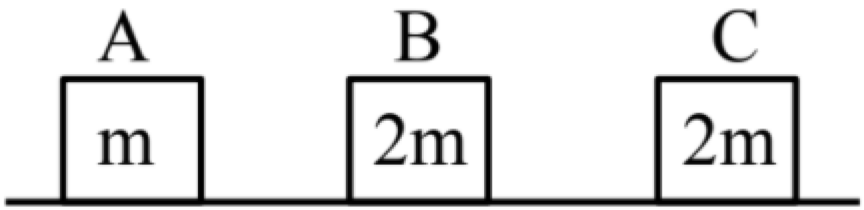
$$(a)\,\mathrm{I}_1=\frac{\mathrm{ML}^2}{12}$$

$$(b)\,\mathrm{I}_2=\frac{(2\mathrm{M})\mathrm{L}^2}{3}=\frac{2\mathrm{ML}^2}{3}$$

$$(c)\,\mathrm{I}_3=\frac{(\mathrm{M})(2\ \mathrm{L})^2}{12}=\frac{\mathrm{ML}^2}{3}$$

$$(d)\,I_4=\frac{(2M)(2L)^2}{3}=\frac{8}{3}ML^2$$

Q5: Three objects A, B and C are kept in a straight line on a frictionless horizontal surface. The masses of A, B and C are m, 2m and 2m respectively. A moves towards B with a speed of 9 m/s and makes an elastic collision with it. Thereafter B makes a completely inelastic collision with C. All motions occur along same straight line. The final speed of C is:



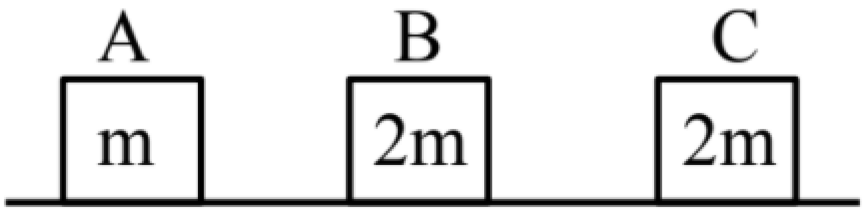
(A) 6 m/s

(B) 9 m/s

(C) 4 m/s

(D) 3 m/s

Q5: Three objects A, B and C are kept in a straight line on a frictionless horizontal surface. The masses of A, B and C are m, 2m and 2m respectively. A moves towards B with a speed of 9 m/s and makes an elastic collision with it. Thereafter B makes a completely inelastic collision with C. All motions occur along same straight line. The final speed of C is:



(A) 6 m/s

(B) 9 m/s

(C) 4 m/s

(D) 3 m/s

Solution:

After 1 st collision

$$\mathbf{V_B = \frac{2\ V_0}{3}}$$

After inelastic collision

$$\mathbf{V_C = \frac{V_B}{2}}$$

Q6: A capacitor of capacitance $C = 1 \ \mu F$ is suddenly connected to a battery of 100 volt through a resistance $R = 100\,\Omega$.The time taken for the capacitor to be charged to get 50 V is:
[Take In 2 = 0.69]

(A) $1.44 \times 10^{-4} s$

(B) $3.33 \times 10^{-4} s$

(C) $0.69 \times 10^{-4} s$

(D) $0.30 \times 10^{-4} s$

Q6: A capacitor of capacitance $C = 1 \ \mu F$ is suddenly connected to a battery of 100 volt through a resistance $R = 100\,\Omega$.The time taken for the capacitor to be charged to get 50 V is:
[Take In 2 = 0.69]

(A) $1.44 \times 10^{-4} s$

(B) $3.33 \times 10^{-4} s$

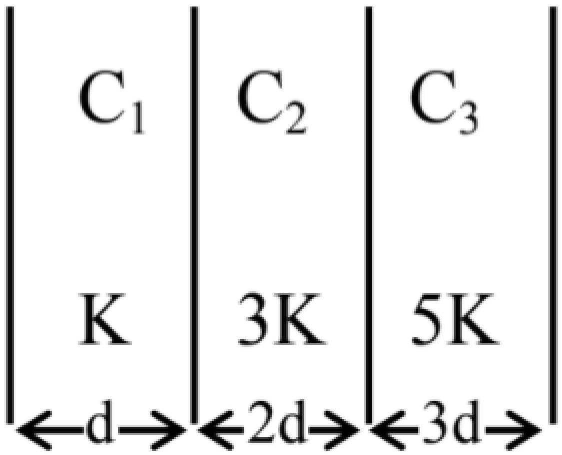
(C) $0.69 \times 10^{-4} s$

(D) $0.30 \times 10^{-4} s$

Solution:

$$50 = 100\left(1 - e^{-\frac{t}{RC}}\right)$$
$$\Rightarrow t = RC \ln 2$$
$$= 100 \times 10^{-6} \times (0.69)$$
$$= 0.69 \times 10^{-4} \text{ s}$$

Q7: In the reported figure, a capacitor is formed by placing a compound dielectric between the plates of parallel plate capacitor. The expression for the capacity of the said capacitor will be: (Given area of plate = A)



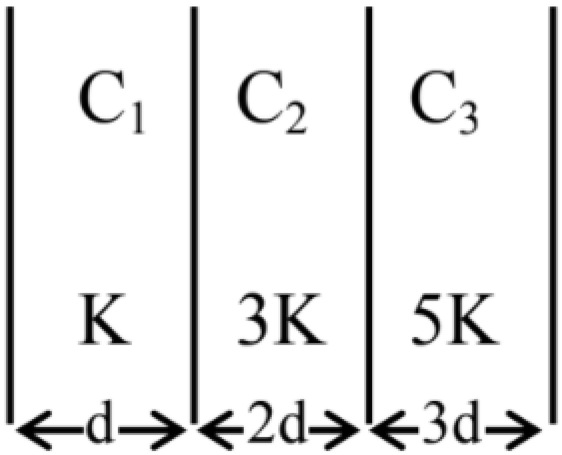
(A) $\frac{15}{34}\frac{K\varepsilon_0 A}{d}$

(B) $\frac{15}{6}\frac{K\varepsilon_0 A}{d}$

(C) $\frac{25}{6}\frac{K\varepsilon_0 A}{d}$

(D) $\frac{9}{6}\frac{K\varepsilon_0 A}{d}$

Q7: In the reported figure, a capacitor is formed by placing a compound dielectric between the plates of parallel plate capacitor. The expression for the capacity of the said capacitor will be: (Given area of plate = A)



(A) $\frac{15}{34}\frac{K\varepsilon_0 A}{d}$

(B) $\frac{15}{6}\frac{K\varepsilon_0 A}{d}$

(C) $\frac{25}{6}\frac{K\varepsilon_0 A}{d}$

(D) $\frac{9}{6} \frac{K\varepsilon_0 A}{d}$

Solution:

$$C_1 = \frac{K\varepsilon_0 A}{d}$$

$$C_2 = \frac{3K\varepsilon_0 A}{2d}$$

$$C_3 = \frac{5K\varepsilon_0 A}{3d}$$

$$\frac{1}{C_{\mathrm{eq}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

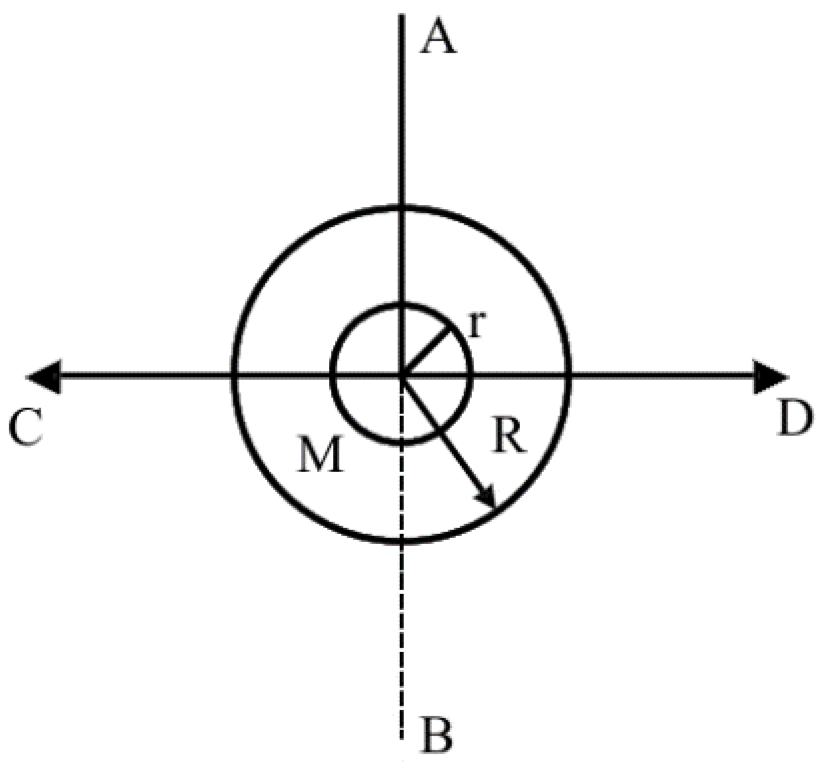
$$\frac{1}{C_{eq}} = \frac{d}{K\varepsilon_0 A} + \frac{2d}{3K\varepsilon_0 A} + \frac{3d}{5K\varepsilon_0 A}$$

$$\frac{1}{C_{eq}} = \frac{d}{K\varepsilon_0 A}\left[1 + \frac{2}{3} + \frac{3}{5}\right]$$

$$= \frac{d}{15K\varepsilon_0 A}[15 + 10 + 9]$$

$$C_{eq} = \frac{15K\varepsilon_0 A}{34d}$$

Q8: The figure shows two solid discs with radius R and r respectively. If mass per unit area is same for both, what is the ratio of MI of bigger disc around axis AB (Which is $\perp$ to the plane of the disc and passing through its centre) of MI of smaller disc around one of its diameters lying on its plane? Given 'M' is the mass of the larger disc. (MI stands for moment of inertia)



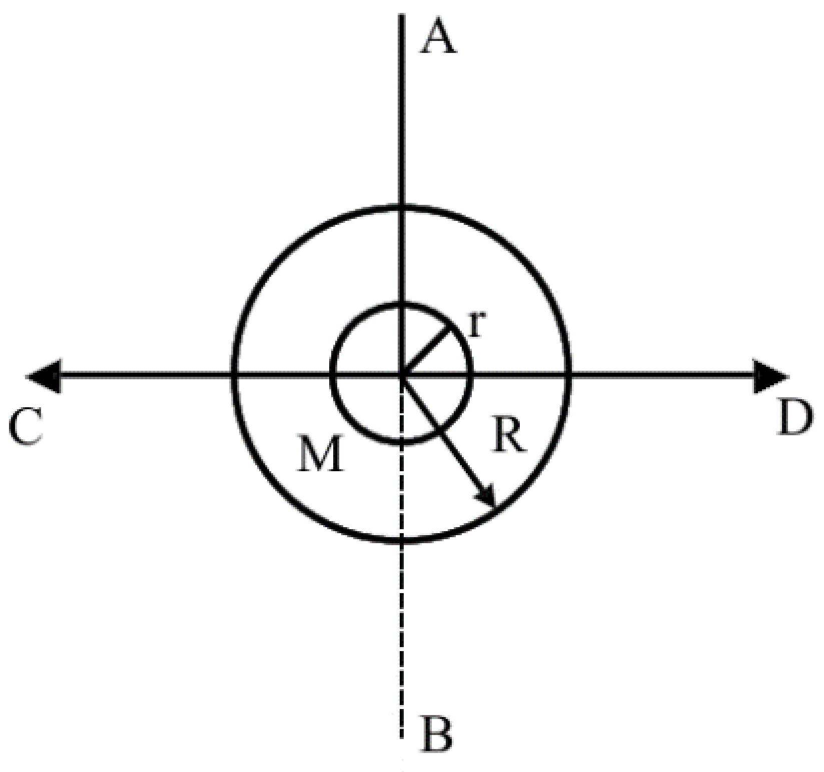
(A) $R^2 : r^2$

(B) $2r^4 : R^4$

(C) $2R^2 : r^2$

(D) $2R^4 : r^4$

Q8: The figure shows two solid discs with radius R and r respectively. If mass per unit area is same for both, what is the ratio of MI of bigger disc around axis AB (Which is $\perp$ to the plane of the disc and passing through its centre) of MI of smaller disc around one of its diameters lying on its plane? Given 'M' is the mass of the larger disc. (MI stands for moment of inertia)



(A) $R^2 : r^2$

(B) $2r^4 : R^4$

(C) $2R^2 : r^2$

(D) $2R^4 : r^4$

Solution:

$$I_1 = \frac{MR^2}{2}$$

$$I_2 = \frac{mr^2}{4}$$

$$m = (\rho)\pi r^2 = \frac{m}{\pi R^2}\pi r^2 = \frac{Mr^2}{R^2}$$

$$\Rightarrow \frac{I_1}{I_2} = \frac{2R^4}{r^4}$$

Q9: In Young's double slit experiment, if the source of light changes from orange to blue then:

(A) the central bright fringe will become a dark fringe.

(B) the distance between consecutive fringes will decrease.

(C) the distance between consecutive fringes will increase.

(D) the intensity of the minima will increase.

Q9: In Young's double slit experiment, if the source of light changes from orange to blue then:

(A) the central bright fringe will become a dark fringe.

(B) the distance between consecutive fringes will decrease.

(C) the distance between consecutive fringes will increase.

(D) the intensity of the minima will increase.

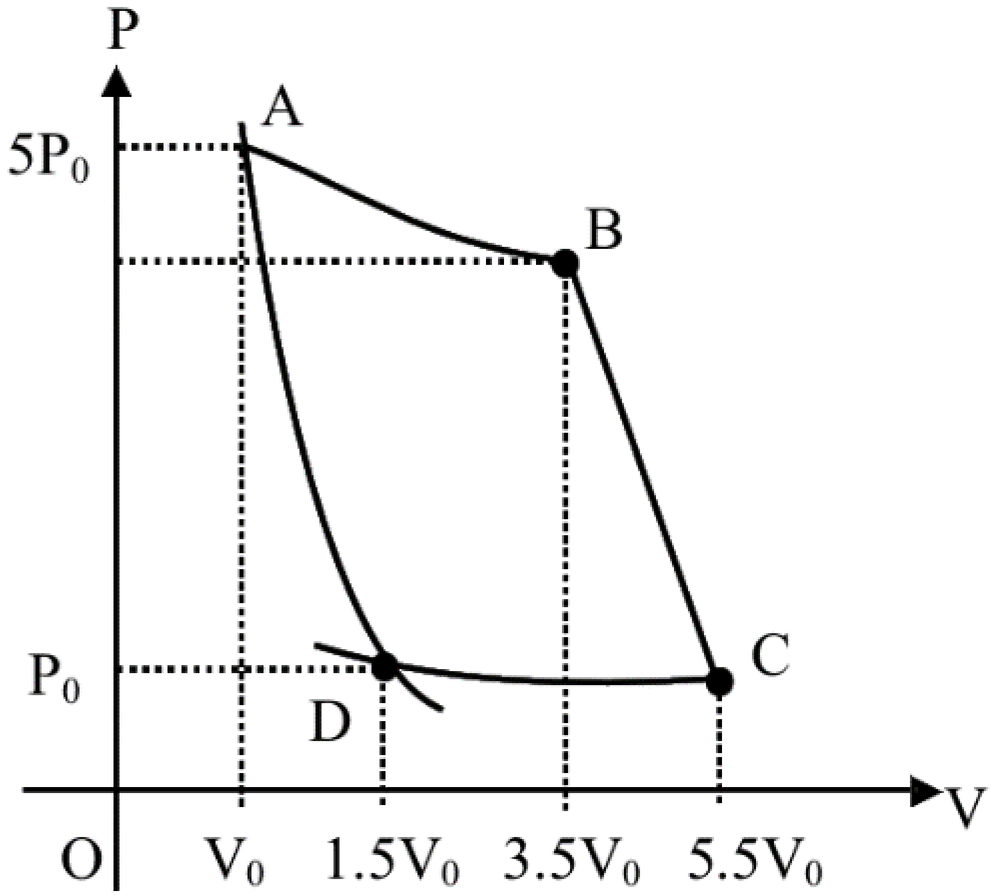
Solution:

Sol. $\lambda_0 > \lambda_B$

Fringe Width, $\beta = \frac{\lambda \mathrm{D}}{d}$

$\Rightarrow \beta_{\mathrm{O}} > \beta_{\text{Blue}}$

Q10: In the reported figure, there is a cyclic process ABCDA on a sample of 1 mol of a diatomic gas. The temperature of the gas during the process A $\rightarrow$ B and C $\rightarrow$ D are T_1 and T_2 $(T_1 > T_2)$respectively.



Choose the correct option out of the following for work done if processes BC and DA are adiabatic.

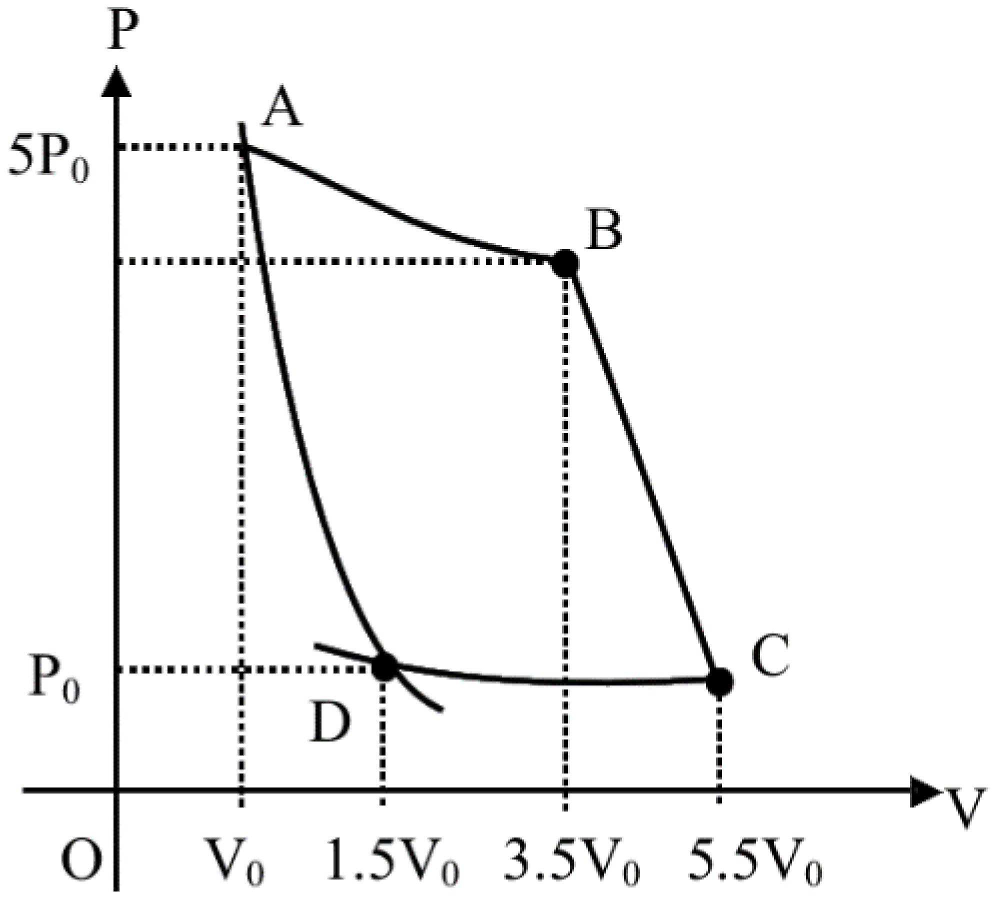
(A) $W_{AB} = W_{DC}$

(B) $W_{AD} = W_{BC}$

(C) $W_{BC} + W_{DA} > 0$

(D) $W_{AB} < W_{CD}$

Q10: In the reported figure, there is a cyclic process ABCDA on a sample of 1 mol of a diatomic gas. The temperature of the gas during the process A $\rightarrow$ B and C $\rightarrow$ D are $T_1 \, and \, T_2 \, (T_1 > T_2)$respectively.



Choose the correct option out of the following for work done if processes BC and DA are adiabatic.

(A) $W_{AB} = W_{DC}$

(B) $W_{AD} = W_{BC}$

(C) $W_{BC} + W_{DA} > 0$

(D) $W_{AB} < W_{CD}$

Solution:

$$\mathrm{W}_{\mathrm{A} \rightarrow \mathrm{D}}=\frac{5 \mathrm{P}_0 \mathrm{~V}_0-\left(\mathrm{P}_0\right)\left(1.5 \mathrm{~V}_0\right)}{\gamma-1}$$

$$\mathrm{W}_{\mathrm{BC}}=\frac{\left(\mathrm{P}_{\mathrm{B}}\right)\left(3.5 \mathrm{~V}_0\right)-\mathrm{P}_{\mathrm{C}}\left(5.5 \mathrm{~V}_0\right)}{\gamma-1}$$

$$\mathrm{P}_{\mathrm{B}}=\frac{5 \mathrm{P}_0}{3.5}=\frac{10}{7} \mathrm{P}_0, \mathrm{P}_{\mathrm{C}}=\frac{1.5 \mathrm{P}_0}{5.5}=\frac{3}{11} \mathrm{P}_0$$

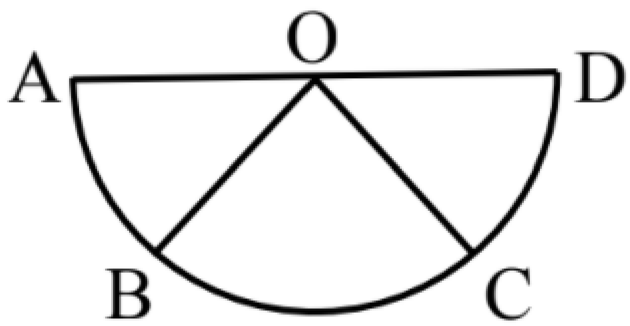
$$\Rightarrow \quad \mathrm{W}_{\mathrm{BC}}=\frac{5 \mathrm{P}_0 \mathrm{~V}_0-1.5 \mathrm{P}_0 \mathrm{~V}_0}{\gamma-1}$$

$$\Rightarrow \quad \mathrm{W}_{\mathrm{AD}}=\mathrm{W}_{\mathrm{BC}}$$

Q11: **Assertion A:** If A, B, C, D are four points on a semi – circular arc with centre at 'O' such that

$\left|\overrightarrow{AB}\right| = \left|\overrightarrow{BC}\right| = \left|\overrightarrow{CD}\right|$,and $\overrightarrow{AB} + \overrightarrow{AC} + \overrightarrow{AD} = 4\overrightarrow{AO} + \overrightarrow{OB} + \overrightarrow{OC}$

Reason R: Polygon law of vector addition yields $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD} + \overrightarrow{AD} = 4\overrightarrow{AO}$



In the light of the above statements, choose the most appropriate answer from the options given below:

(A) A is correct but R is not correct.

(B) A is not correct but R is correct.

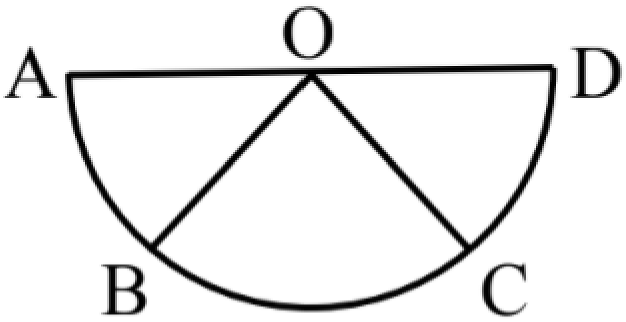
(C) Both A and R are correct and R is the correct explanation of A.

(D) Both A and R are correct but R is not the correct explanation of A.

Q11: **Assertion A:** If A, B, C, D are four points on a semi – circular arc with centre at 'O' such that

$\left|\overrightarrow{AB}\right| = \left|\overrightarrow{BC}\right| = \left|\overrightarrow{CD}\right|$,and $\overrightarrow{AB} + \overrightarrow{AC} + \overrightarrow{AD} = 4\overrightarrow{AO} + \overrightarrow{OB} + \overrightarrow{OC}$

Reason R: Polygon law of vector addition yields $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD} + \overrightarrow{AD} = 4\overrightarrow{AO}$



In the light of the above statements, choose the most appropriate answer from the options given below:

(A) A is correct but R is not correct.

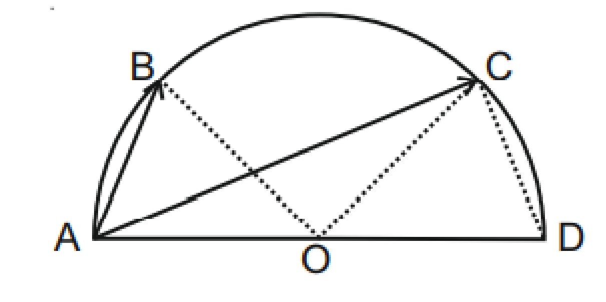
(B) A is not correct but R is correct.

(C) Both A and R are correct and R is the correct explanation of A.

(D) Both A and R are correct but R is not the correct explanation of A.

Solution:

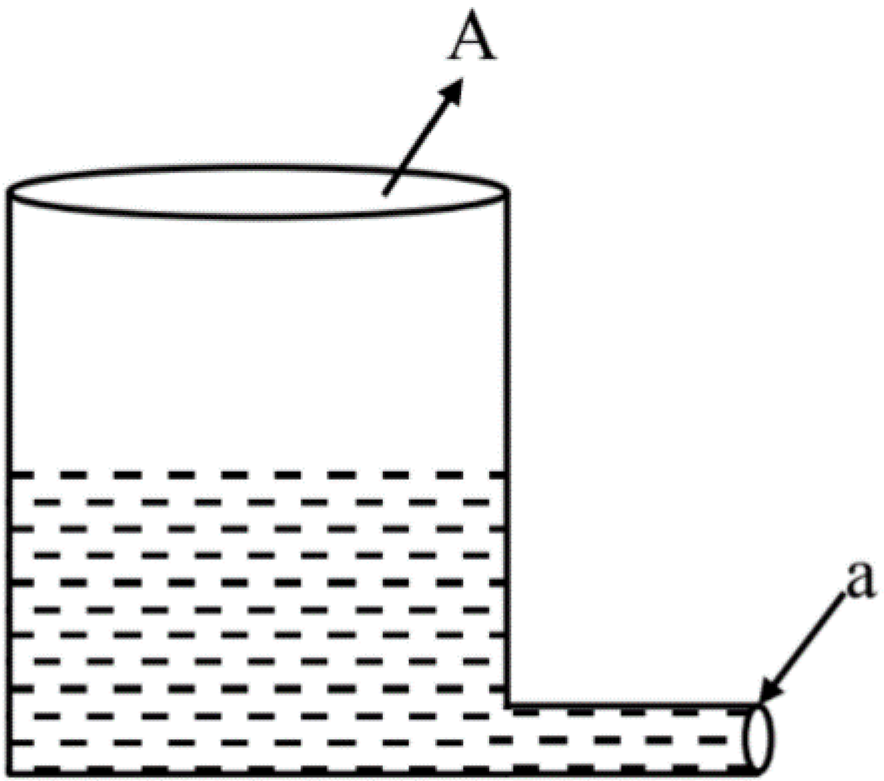
$\overline{AB} + \overline{AC} + \overline{AD}$



$= (\overline{AO} + \overline{OB}) + (\overline{AO} + \overline{OC}) + 2\overline{AO}$

$= 4\overline{AO} + \overline{OB} + \overline{OC}$

Q12: A light cylindrical vessel is kept on a horizontal surface. Area of base is A. A hole of cross-sectional area 'a' is made just at its bottom side. The minimum coefficient of friction necessary to prevent sliding the vessel due to the impact force of the emerging liquid is (a < < A)



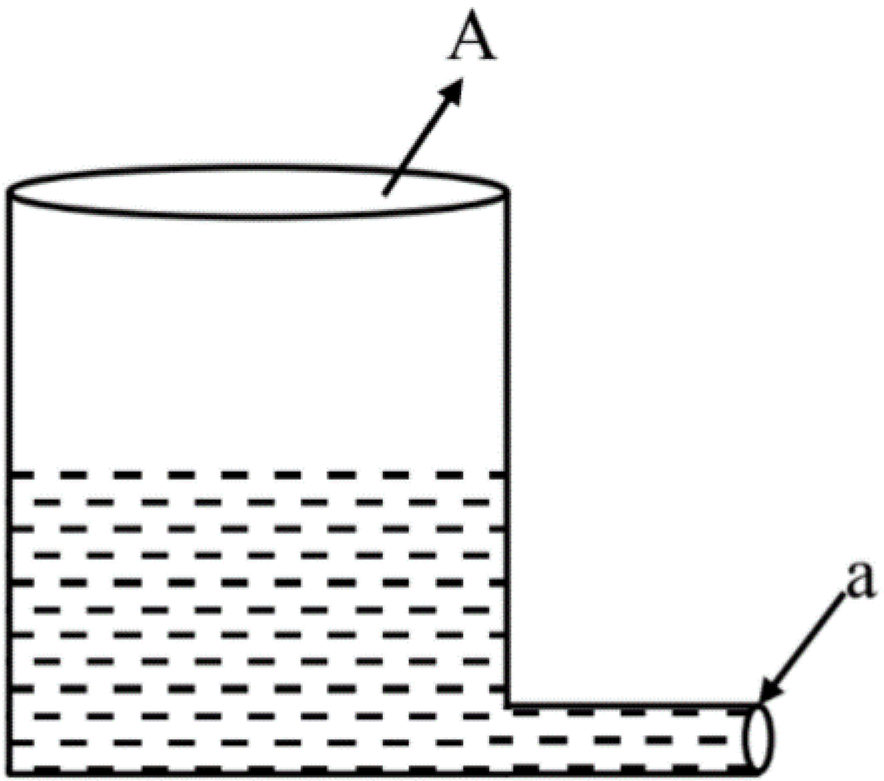
(A) $\frac{A}{2a}$

(B) None of these

(C) $\frac{2a}{A}$

(D) $\frac{a}{A}$

Q12: A light cylindrical vessel is kept on a horizontal surface. Area of base is A. A hole of cross-sectional area 'a' is made just at its bottom side. The minimum coefficient of friction necessary to prevent sliding the vessel due to the impact force of the emerging liquid is ($a << A$)



(A) $\frac{A}{2a}$

(B) None of these

(C) $\frac{2a}{A}$

(D) $\frac{a}{A}$

Solution:

$$V_e = \sqrt{2gh}$$

$$\text{Thrust force } = \text{ friction}$$

$$\rho a v^2 = \mu(\rho \mathrm{Ah})\mathrm{g}$$

$$a(2gh) = \mu A g h$$

$$\mu = \frac{2a}{\mathrm{A}}$$

Q13: A particle starts executing simple harmonic motion (SHM) of amplitude ‘a’ and total energy E at any instant, its kinetic energy is $\frac{3E}{4}$ then its displacement ‘y' is given by:

(A) $y = a$

(B) $y = \frac{a}{\sqrt{2}}$

(C) $y = \frac{a\sqrt{3}}{2}$

(D) $y = \frac{a}{2}$

Q13: A particle starts executing simple harmonic motion (SHM) of amplitude ‘a’ and total energy E at any instant, its kinetic energy is $\frac{3E}{4}$ then its displacement ‘y' is given by:

(A) $y = a$

(B) $y = \frac{a}{\sqrt{2}}$

(C) $y = \frac{a\sqrt{3}}{2}$

(D) $y = \frac{a}{2}$

Solution:

$$E = \frac{1}{2}m\omega^2 a^2$$

$$\text{K.E.} = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2\left[a^2 - y^2\right]$$

$$a^2 - y^2 = \frac{3}{4}a^2$$

$$y = \frac{a}{2}$$

Q14: If 'f' denotes the ratio of the number of nuclei decayed $\left(N_d\right)$ to the number of nuclei at $t=0\left(N_0\right)$ then for a collection of radioactive nuclei, the rate of change of 'f' with respect to time is given as: [λ is the radioactive decay constant]

(A) $-\lambda\left(1-e^{-\lambda t}\right)$

(B) $\lambda\left(1-e^{-\lambda t}\right)$

(C) $\lambda e^{-\lambda t}$

(D) $-\lambda e^{-\lambda t}$

Q14: If 'f' denotes the ratio of the number of nuclei decayed $\left(N_d\right)$ to the number of nuclei at $t=0\left(N_0\right)$ then for a collection of radioactive nuclei, the rate of change of 'f' with respect to time is given as: [λ is the radioactive decay constant]

(A) $-\lambda\left(1-e^{-\lambda t}\right)$

(B) $\lambda\left(1-e^{-\lambda t}\right)$

(C) $\lambda e^{-\lambda t}$

(D) $-\lambda e^{-\lambda t}$

Solution:

$$N=N_0 e^{-\lambda t}$$

$$\mathrm{N}_d=\mathrm{N}_0-\mathrm{N}$$

$$\mathrm{N}_d=\mathrm{N}_0\left(1-e^{-\lambda t}\right)$$

$$f=\frac{\mathrm{N}_d}{\mathrm{N}_0}=1-e^{-\lambda t}$$

$$\frac{df}{dt}=0-(-\lambda)e^{-\lambda t}=\lambda e^{-\lambda t}$$

Q15: Two capacitors of capacities 2C and C are joined in parallel and charged up to potential V. The battery is removed and the capacitor of capacity C is filled completely with a medium of dielectric constant K. The potential difference across the capacitors will now be:

(A) $\frac{V}{K+2}$

(B) $\frac{V}{K}$

(C) $\frac{3V}{K+2}$

(D) $\frac{3V}{K}$

Q15: Two capacitors of capacities 2C and C are joined in parallel and charged up to potential V. The battery is removed and the capacitor of capacity C is filled completely with a medium of dielectric constant K. The potential difference across the capacitors will now be:

(A) $\frac{V}{K+2}$

(B) $\frac{V}{K}$

(C) $\frac{3V}{K+2}$

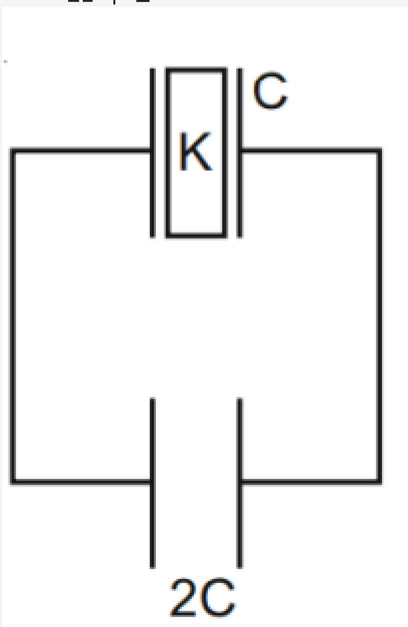
(D) $\frac{3V}{K}$

Solution:

$$Q_{\text{total}} = 3C \times V$$

$$\therefore V_f = \frac{3CV}{(2C + KC)}$$

$$= \frac{3}{K+2} \times V$$



Q16: A ball is thrown up with a certain velocity so that it reaches a height ‘h’. find the ratio of the two different times of the ball reaching $\frac{h}{3}$ in both the directions.

(A) $\frac{\sqrt{2}-1}{\sqrt{2}+1}$

(B) $\frac{1}{3}$

(C) $\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$

(D) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$

Q16: A ball is thrown up with a certain velocity so that it reaches a height ‘h’. find the ratio of the two different times of the ball reaching $\frac{h}{3}$ in both the directions.

(A) $\frac{\sqrt{2}-1}{\sqrt{2}+1}$

(B) $\frac{1}{3}$

(C) $\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$

(D) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$

Solution:

$$u=\sqrt{2gh}$$

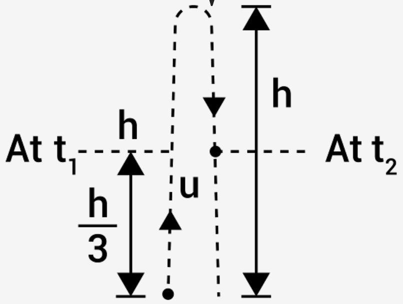
Now using, $s=ut+\frac{1}{2}at^2$

$$\frac{h}{3}=ut-\frac{1}{2}gt^2$$

$$\frac{1}{2}gt^2-ut+\frac{4}{3}=0$$

$$t=\frac{u\pm\sqrt{u^2-4\cdot\frac{1}{2}g\frac{h}{3}}}{g}$$

$$\frac{t_1}{t_2}=\frac{\sqrt{2}-\sqrt{\frac{4}{3}}}{\sqrt{2}+\sqrt{\frac{4}{3}}}=\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$$



Q17: A 0.07 H inductor and a $12\ \Omega$ resistor are connected in series to a 220 V, 50 Hz ac source. The approximate current in the circuit and the phase angle between current and source voltage are respectively. [Take $\pi\ as\ \frac{22}{7}$]

(A) $8.8\ A\ and\ \tan^{-1}\left(\frac{11}{6}\right)$

(B) $88\ A\ and\ \tan^{-1}\left(\frac{11}{6}\right)$

(C) $0.88\ A\ and\ \tan^{-1}\left(\frac{11}{6}\right)$

(D) $8.8\ A\ and\ \tan^{-1}\left(\frac{6}{11}\right)$

Q17: A 0.07 H inductor and a $12\ \Omega$ resistor are connected in series to a 220 V, 50 Hz ac source. The approximate current in the circuit and the phase angle between current and source voltage are respectively. [Take $\pi\ as\ \frac{22}{7}$]

(A) $8.8\ A\ and\ \tan^{-1}\left(\frac{11}{6}\right)$

(B) $88\ A\ and\ \tan^{-1}\left(\frac{11}{6}\right)$

(C) $0.88\ A\ and\ \tan^{-1}\left(\frac{11}{6}\right)$

(D) $8.8\ A\ and\ \tan^{-1}\left(\frac{6}{11}\right)$

Solution:

$$\mathrm{Z}=\sqrt{12^2+(100\pi\times0.07)^2}\approx25\Omega$$

$$\therefore \mathrm{I}=\frac{220}{25}=8.8\ \mathrm{A}$$

$$\tan\phi=\frac{\mathrm{X_L}}{\mathrm{R}}=\frac{100\times\frac{22}{7}\times\frac{7}{100}}{12}=\frac{11}{6}$$

Q18: Two identical tennis balls each having mass 'm' and charge 'q' are suspended from a fixed point by threads of length 'l'. What is the equilibrium separation when each thread makes a small angle $'\theta'$ with the vertical?

(A) $x = \left(\frac{q^2 l}{2\pi\varepsilon_0 mg}\right)^{\frac{1}{2}}$

(B) $x = \left(\frac{q^2 l}{2\pi\varepsilon_0 mg}\right)^{\frac{1}{3}}$

(C) $x = \left(\frac{q^2 l^2}{2\pi\varepsilon_0 m^2 g}\right)^{\frac{1}{3}}$

(D) $x = \left(\frac{q^2 l^2}{2\pi\varepsilon_0 m^2 g^2}\right)^{\frac{1}{3}}$

Q18: Two identical tennis balls each having mass 'm' and charge 'q' are suspended from a fixed point by threads of length 'l'. What is the equilibrium separation when each thread makes a small angle $'\theta'$ with the vertical?

(A) $x = \left(\frac{q^2 l}{2\pi\varepsilon_0 mg}\right)^{\frac{1}{2}}$

(B) $x = \left(\frac{q^2 l}{2\pi\varepsilon_0 mg}\right)^{\frac{1}{3}}$

(C) $x = \left(\frac{q^2 l^2}{2\pi\varepsilon_0 m^2 g}\right)^{\frac{1}{3}}$

(D) $x = \left(\frac{q^2 l^2}{2\pi\varepsilon_0 m^2 g^2}\right)^{\frac{1}{3}}$

Solution:

$$\mathrm{T}\cos\theta = mg$$

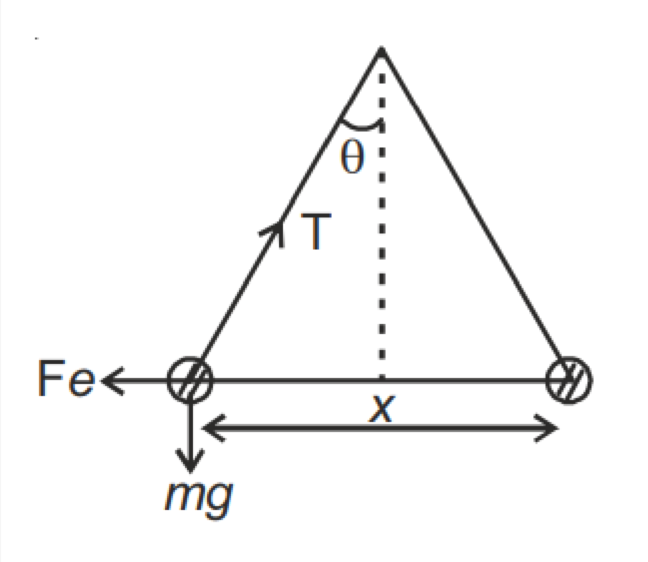
$$\mathrm{T}\sin\theta = \mathrm{F}_e$$

$$\tan\theta = \frac{\mathrm{F}_e}{mg}$$

$$\frac{x}{2l} = \frac{q^2}{4\pi\varepsilon_0 x^2 mg}$$

$$x^3 = \frac{q^2 l}{2\pi\varepsilon_0 mg}$$

$$x = \left(\frac{q^2 l}{2\pi\varepsilon_0 mg}\right)^{\frac{1}{3}}$$



Q19: **Assertion A:** If in five complete rotations of the circular scale, the distance travelled on main scale of the screw gauge is 5 mm and there are 50 total divisions on circular scale, then least count is 0.001 cm.

Reason R: $Least\ Count = \frac{Pitch}{Total\ divisions\ on\ circular\ scale}$

In the light of the above statements, choose the most appropriate answer from the options given
below:

(A) A is not correct but R is correct.

(B) Both A and R are correct and R is the correct explanation of A.

(C) A is correct but R is not correct.

(D) Both A and R are correct and R is NOT the correct explanation of A.

Q19: **Assertion A:** If in five complete rotations of the circular scale, the distance travelled on main scale of the screw gauge is 5 mm and there are 50 total divisions on circular scale, then least count is 0.001 cm.

Reason R: $Least\ Count = \frac{Pitch}{Total\ divisions\ on\ circular\ scale}$

In the light of the above statements, choose the most appropriate answer from the options given
below:

(A) A is not correct but R is correct.

(B) Both A and R are correct and R is the correct explanation of A.

(C) A is correct but R is not correct.

(D) Both A and R are correct and R is NOT the correct explanation of A.

Solution:

Pitch $= \frac{5 \text{ mm}}{5} = 1 \text{ mm}$

$$\text{So least count } = \frac{\text{Pitch}}{\text{Total division on circular scale}}$$

$$= \frac{1 \text{ mm}}{50}$$

$$= 0.02 \text{ mm}$$

$$= 0.002 \text{ cm}$$

So A is not correct but R is correct.

Q20: A body takes 4 min. to cool from $61^oC\ to\ 59^oC$. If the temperature of the surroundings is 30^oC, the time taken by the body to cool from $51^oC\ to\, 49^oC$ is:

(A) 4 min.

(B) 3 min.

(C) 8 min.

(D) 6 min.

Q20: A body takes 4 min. to cool from $61^{o}C\ to\ 59^{o}C$. If the temperature of the surroundings is $30^{o}C$, the time taken by the body to cool from $51^{o}C\ to\,49^{o}C$is:

(A) 4 min.

(B) 3 min.

(C) 8 min.

(D) 6 min.

Solution:

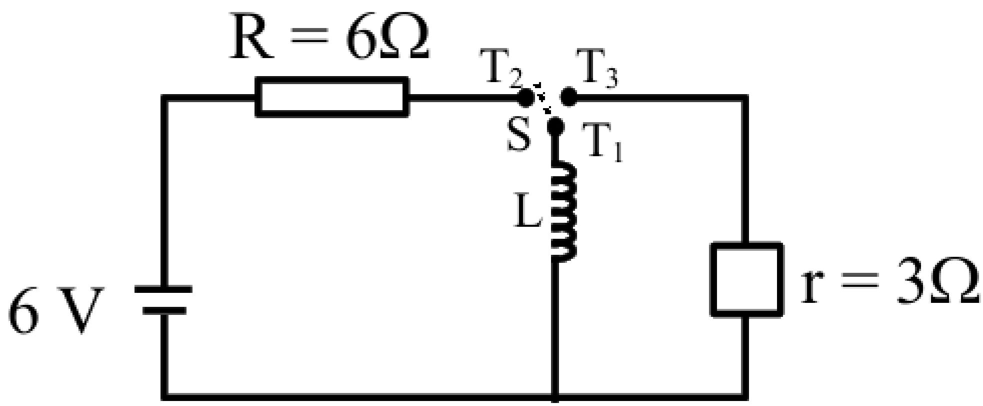
$$-\frac{dT}{dt}=\mathrm{k}\left(\mathrm{T}-\mathrm{T}_{0}\right)$$

$$\frac{(60-30)}{(50-30)}=\frac{t}{4}$$

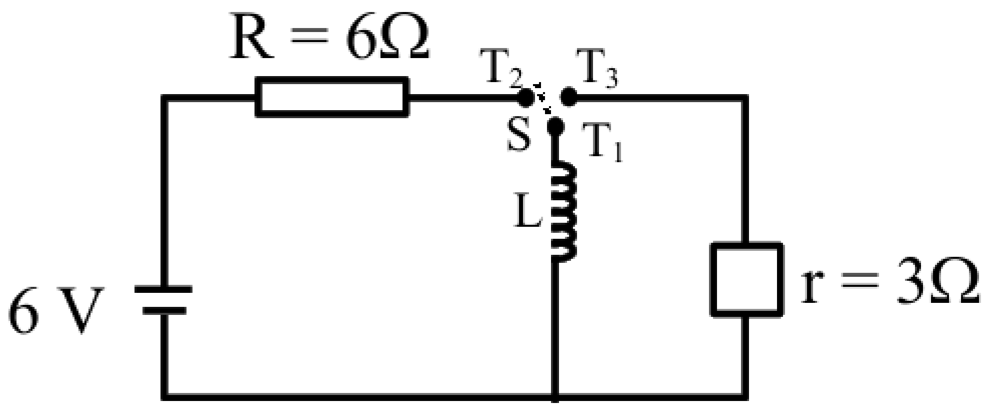
$$t=6\ \text{min.}$$

Q21: Consider an electrical circuit containing a two way switch 'S'. Initially S is open and then T_1 is connected to T_2.As the current in $R = 6\,\Omega$ attains a maximum value of steady state level, T_1 is disconnected from T_2 and immediately connected to T_3. Potential drop across $r = 3\,\Omega$ resistor immediately after T_1 is connected to T_3 is ______ V.

(Round off to the Nearest Integer)



Q21: Consider an electrical circuit containing a two way switch 'S'. Initially S is open and then T_1 is connected to T_2.As the current in $R = 6\,\Omega$ attains a maximum value of steady state level, T_1 is disconnected from T_2 and immediately connected to T_3. Potential drop across $r = 3\,\Omega$ resistor immediately after T_1 is connected to T_3 is ______ V.
(Round off to the Nearest Integer)



3.00

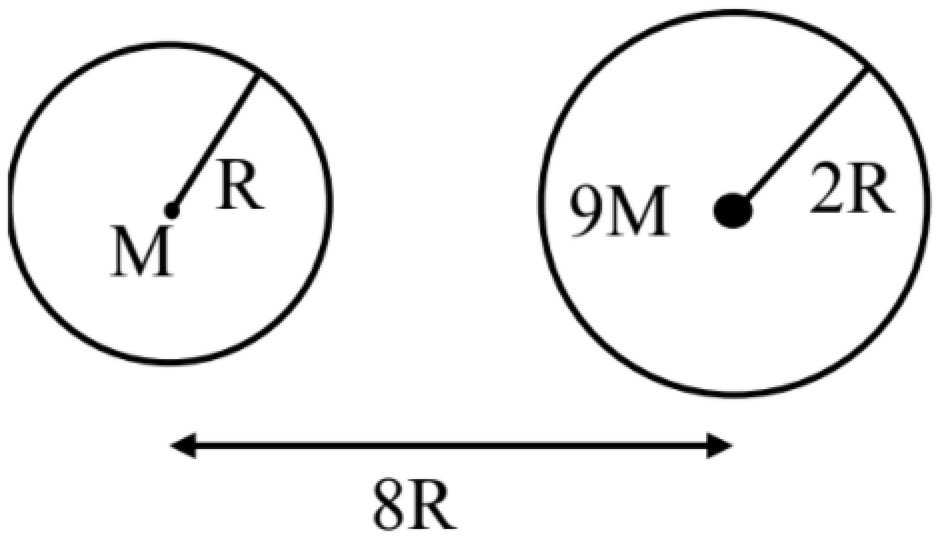
Solution:

Sol. $i_0 = \frac{\varepsilon}{R} = \frac{6}{6} = 1 \mathrm{~A}$

As current through inductor will not change instantly $\mathrm{V}_r = i_0 r = 1 \times 3 = 3 \mathrm{~V}$

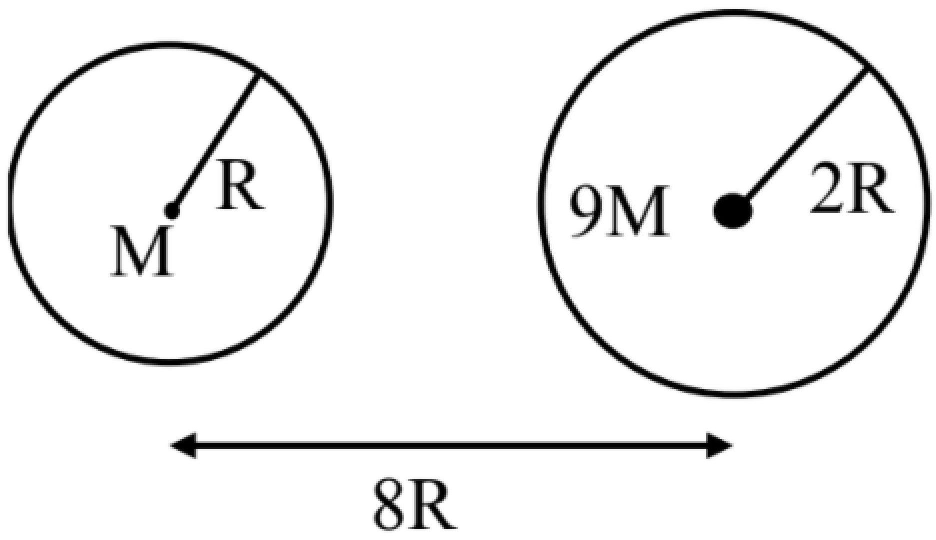
Q22: Suppose two planets (spherical in shape) of radii R and 2R, but mass M and 9 M respectively have a centre to centre separation 8 R as shown in the figure. A satellite of mass 'm' is projected from the surface of the planet of mass 'M' directly towards the centre of the second planet. The minimum speed 'v' required for the satellite to reach the surface of the second planet is $\sqrt{\frac{a}{7}\frac{GM}{R}}$ then the value of 'a' is ________.

[Given: The two planets are fixed in their position]



Q22: Suppose two planets (spherical in shape) of radii R and 2R, but mass M and 9 M respectively have a centre to centre separation 8 R as shown in the figure. A satellite of mass 'm' is projected from the surface of the planet of mass 'M' directly towards the centre of the second planet. The minimum speed 'v' required for the satellite to reach the surface of the second planet is $\sqrt{\frac{a}{7}\frac{GM}{R}}$ then the value of 'a' is ________.

[Given: The two planets are fixed in their position]



4.00

Solution:

$$-\frac{\mathrm{GM}m}{2\mathrm{R}} - \frac{9\mathrm{GMm}}{6\mathrm{R}} = \frac{1}{2}mv^2 - \frac{\mathrm{GMm}}{\mathrm{R}} - \frac{9\mathrm{GMm}}{7\mathrm{R}}$$

$$\frac{1}{2}mv^2 = \frac{16\mathrm{GM}}{7\mathrm{R}} - \frac{2\mathrm{GM}}{\mathrm{R}}$$

$$\frac{1}{2}mv^2 = \frac{2\mathrm{GM}}{7\mathrm{R}}$$

$$v = \sqrt{\frac{4\mathrm{GM}}{7\mathrm{R}}}$$

Q23: In Bohr's atomic model, the electron is assumed to revolve in a circular orbit of radius $0.5\ \overset{o}{A}$. If the speed of electron is $2.2 \times 10^6\ m/s$, then the current associated with the electron will be ________$\times 10^{-2} mA$. $[Take\ \pi\ as \frac{22}{7}]$

Q23: In Bohr's atomic model, the electron is assumed to revolve in a circular orbit of radius $0.5\ \overset{o}{A}$. If the speed of electron is $2.2 \times 10^6\ m/s$, then the current associated with the electron will be ________ $\times 10^{-2} mA$. $[Take\ \pi\ as \frac{22}{7}]$

112.00

Solution:

$$i = \frac{qv}{2\pi r}$$
$$= \frac{1.6 \times 10^{-19} \times 2.2 \times 10^6}{2 \times \pi \times 0.5 \times 10^{-10}}$$
$$= 112 \times 10^{-5}\ \mathrm{A}$$

Q24: A radioactive sample has an average life of 30 ms and is decaying. A capacitor of capacitance 200 μF is first charged and later connected with resistor 'R'. If the ratio of charge on capacitor to the activity of radioactive sample is fixed with respect to time then the value of 'R' should be _______ Ω.

Q24: A radioactive sample has an average life of 30 ms and is decaying. A capacitor of capacitance 200 μF is first charged and later connected with resistor 'R'. If the ratio of charge on capacitor to the activity of radioactive sample is fixed with respect to time then the value of 'R' should be ______ Ω.

150.00

Solution:

$$\frac{q}{A}=\frac{Q}{A_0}$$

$$\Rightarrow \frac{1}{RC}=\lambda=\frac{1}{T_{avg}}$$

$$\Rightarrow R=\frac{T_{avg.}}{C}=\frac{30\times10^{-3}}{200\times10^{-6}}$$

$$=150\Omega$$

Q25: A particle of mass $9.1 \times 10^{-31} kg$ travels in a medium with speed of $10^{6}\ m/s$ and a photon of a radiation of linear momentum $10^{-27} kg\, m/s$ travels in vacuum. The wavelength of photon is _______ times the wavelength of the particle.

Q25: A particle of mass $9.1 \times 10^{-31} kg$ travels in a medium with speed of $10^6\ m/s$ and a photon of a radiation of linear momentum $10^{-27} kg\, m/s$ travels in vacuum. The wavelength of photon is _______ times the wavelength of the particle.

910.00

Solution:

$$\frac{\lambda_1}{\lambda_2} = \frac{p_2}{p_1}$$

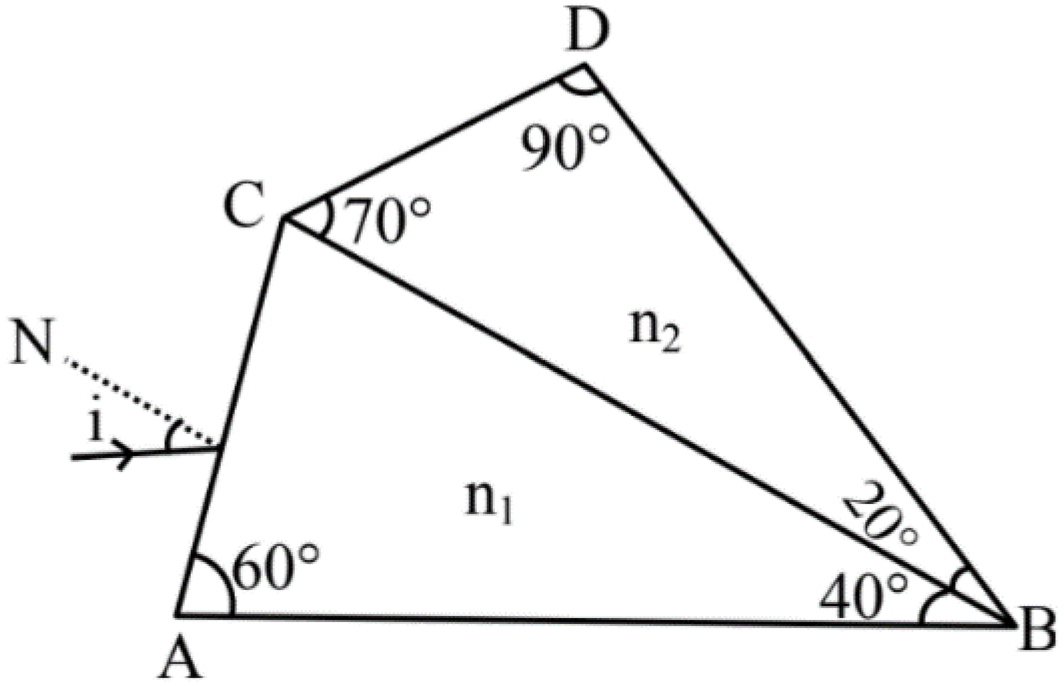
$$\Rightarrow \lambda_2 = \frac{\lambda_1 p_1}{p_2} = \frac{9.1 \times 10^{-31} \times 10^6}{10^{-27}} \lambda_1$$

$$= 910 \lambda_1$$

Q26: A prism of refractive index n_1 and another prism of refractive index n_2 are stuck together (as shown in the figure). n_1 and n_2 depend on λ, the wavelength of light, according to the relation

$$n_1 = 1.2 + \frac{10.8\times10^{-14}}{\lambda^2} \; and \; n_2 = 1.45 + \frac{1.8\times10^{-14}}{\lambda^2}$$

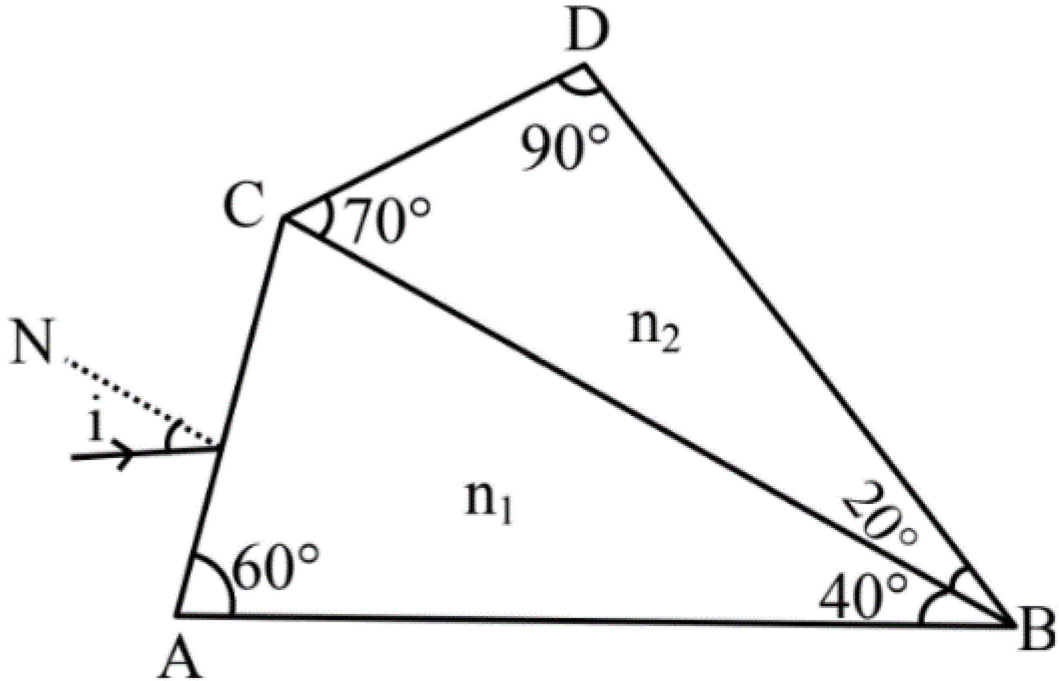
The wavelength for which rays incident at any angle on the interface BC pass through without bending at that interface will be __________ nm.



Q26: A prism of refractive index n_1 and another prism of refractive index n_2 are stuck together (as shown in the figure). n_1 and n_2 depend on λ, the wavelength of light, according to the relation

$$n_1 = 1.2 + \frac{10.8\times10^{-14}}{\lambda^2} \; and \; n_2 = 1.45 + \frac{1.8\times10^{-14}}{\lambda^2}$$

The wavelength for which rays incident at any angle on the interface BC pass through without bending at that interface will be __________ nm.



600.00

Solution:

For no deviation $n_1 = n_2$

$$1.2 + \frac{10.8 \times 10^{-14}}{\lambda^2} = 1.45 + \frac{1.8 \times 10^{-14}}{\lambda^2}$$

$$0.25 = \frac{9 \times 10^{-14}}{\lambda^2}$$

$$\Rightarrow \lambda = \frac{3}{5} \times 10^{-6} = 6 \times 10^{-7}\ \mathrm{m}$$

$$= 600\ \mathrm{nm}$$

Q27: A stone of mass 20 g is projected from a rubber catapult of length 0.1 m and area of cross section $10^{-6} m^2$ stretched by an amount 0.04 m. The velocity of the projected stone is ________m/s.
(Young's modulus of rubber $= 0.5 \times 10^9\ N/m^2$)

Q27: A stone of mass 20 g is projected from a rubber catapult of length 0.1 m and area of cross section $10^{-6} m^2$ stretched by an amount 0.04 m. The velocity of the projected stone is ________m/s.

(Young's modulus of rubber $= 0.5 \times 10^9 \ N/m^2$)

20.00

Solution:

$$\frac{1}{2} y(\text{ strain })^2 \times \text{ volume } = \frac{1}{2} m v^2$$

$$\frac{1}{2} \times 0.5 \times 10^9 \times 16 \times 10^{-2} \times 10^{-7} = \frac{1}{2} \times 20 \times 10^{-3} \times v^2$$

$$\Rightarrow v^2 = 400$$

$$v = 20 \text{ m/s}$$

Q28: A transistor is connected in common emitter circuit configuration, the collector supply voltage is 10 V and the voltage drop across a resistor of 1000 Ω in the collector circuit is 0.6 V. If the current gain factor (β) is 24, then the base current is _______ $\mu A.$ (Round off to the Nearest Integer)

Q28: A transistor is connected in common emitter circuit configuration, the collector supply voltage is 10 V and the voltage drop across a resistor of 1000 Ω in the collector circuit is 0.6 V. If the current gain factor (β) is 24, then the base current is _______ μA. (Round off to the Nearest Integer)

25.00

Solution:

$$(\beta)=\frac{I_C}{I_B} \quad I_C=\frac{0.6}{1000}=6\times10^{-4}\ \mathrm{A}$$

$$24=\frac{6\times10^{-4}\ \mathrm{A}}{x}$$

$$x=\frac{1}{4}\times10^{-4}\ \mathrm{A}=25\times10^{-6}\ \mathrm{A}$$

Q29: The amplitude of upper and lower side bands of A.M. wave where a carrier signal with frequency 11.21 MHz, peak voltage 15 V is amplitude modulated by a 7.7 kHz sine wave of 5V amplitude are $\frac{a}{10} V\ and\ \frac{b}{10}$ respectively, Then the value of $\frac{a}{b}$ is __________.

Q29: The amplitude of upper and lower side bands of A.M. wave where a carrier signal with frequency 11.21 MHz, peak voltage 15 V is amplitude modulated by a 7.7 kHz sine wave of 5V amplitude are $\frac{a}{10}\ V\ and\ \frac{b}{10}$ respectively, Then the value of $\frac{a}{b}$ is __________.

1.00

Solution:

Amplitudes of side band for both left and right side band will be equal.

Ratio of both amplitudes = 1 (as both will be equal)

Q30: In a uniform magnetic field, the magnetic needle has a magnetic moment $9.85 \times 10^{-2}\ A/m^2$ and moment of inertia $5 \times 10^{-6}\ kg\,m^2$. If it performs 10 complete oscillations in 5 seconds then the magnitude of the magnetic field is _____ mT. [Take $\pi^2\ as\ 9.85$]

Q30: In a uniform magnetic field, the magnetic needle has a magnetic moment $9.85 \times 10^{-2}\ A/m^2$ and moment of inertia $5 \times 10^{-6}\ kg\,m^2$. If it performs 10 complete oscillations in 5 seconds then the magnitude of the magnetic field is ______ mT. [Take $\pi^2\ as\ 9.85$]

8.00

Solution:

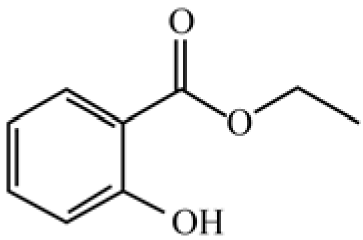
$$\mathrm{T} = 2\pi\sqrt{\frac{I}{\mathrm{MB}}}$$

$$\frac{5}{10} = 2\pi\sqrt{\frac{5 \times 10^{-6}}{9.85 \times 10^{-2} \times \mathrm{B}}}$$

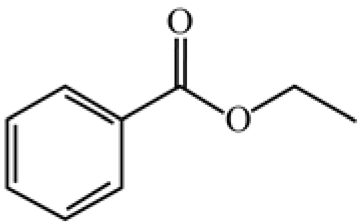
$$\mathrm{B} = 8 \times 10^{-3}\ \mathrm{T}$$

Q31: Which one of the following compounds will give orange precipitate when treated with 2,4-dinitrophenylhydrazine?

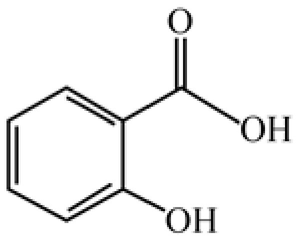
(A)



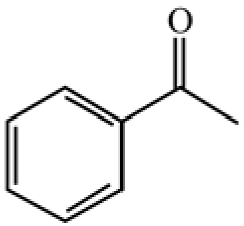
(B)



(C)

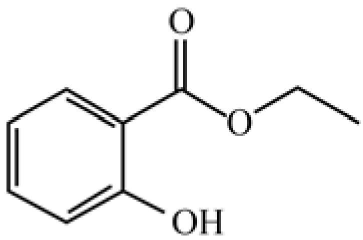


(D)

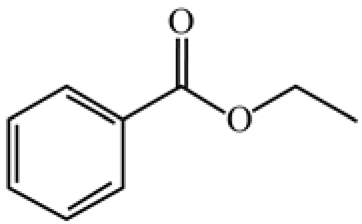


Q31: Which one of the following compounds will give orange precipitate when treated with 2,4-dinitrophenylhydrazine?

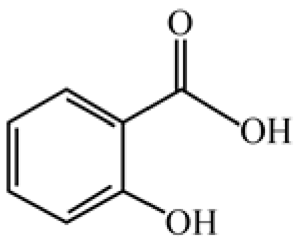
(A)



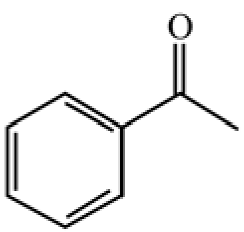
(B)



(C)

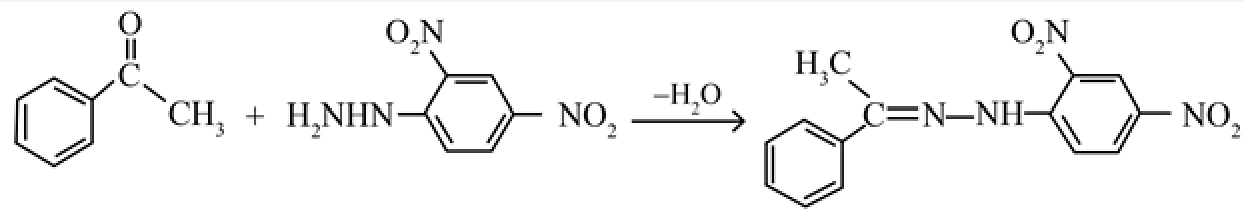


(D)



Solution:

2,4-dinitrophenyl hydrazine solution is used to detect ketones and aldehydes. A positive test is confirmed by the formation of a yellow, orange or red precipitate.



Q32: The product obtained from the electrolytic oxidation of acidified sulphate solutions, is:

(A) HSO^-_4

(B) HO_3SOOSO_3H

(C) HO_2SOSO_2H

(D) HO_2SOSO_3H

Q32: The product obtained from the electrolytic oxidation of acidified sulphate solutions, is:

(A) HSO^{-}_{4}

(B) HO_3SOOSO_3H

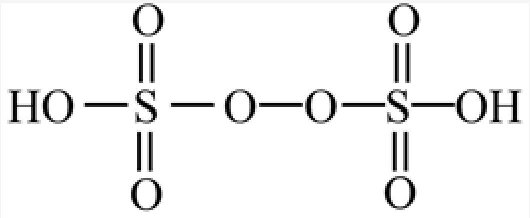
(C) HO_2SOSO_2H

(D) HO_2SOSO_3H

Solution:

$$2SO_4^{2-} \xrightarrow{electrolysis} S_2O_8^{2-} + 2e^-$$

$H_2S_2O_8$:



Q33: The parameters of the unit cell of a substance are a = 2.5, b = 3.0, c = 4.0, $\alpha = \gamma = 90°$, $\beta = 120^o$. The crystal system of the substance is:

(A) Hexagonal

(B) Orthorhombic

(C) Monoclinic

(D) Triclinic

Q33: The parameters of the unit cell of a substance are a = 2.5, b = 3.0, c = 4.0, $\alpha = \gamma = 90°$, $\beta = 120^o$. The crystal system of the substance is:

(A) Hexagonal

(B) Orthorhombic

(C) Monoclinic

(D) Triclinic

Solution:

Given
$\alpha = \gamma = 90°$, $\beta = 120^o$
$a \neq b \neq c$
Thus, crystal system is Monoclinic.

Q34: The oxidation states of 'P' in $H_4P_2O_7$, $H_4P_2O_5$ and $H_4P_2O_6$ respectively, are :

(A) 7, 5 and 6

(B) 5, 4 and 3

(C) 5, 3 and 4

(D) 6, 4 and 5

Q34: The oxidation states of 'P' in $H_4P_2O_7$, $H_4P_2O_5$ and $H_4P_2O_6$ respectively, are :

(A) 7, 5 and 6

(B) 5, 4 and 3

(C) 5, 3 and 4

(D) 6, 4 and 5

Solution:

$H_4P_2O_7$: $4 + 2x - 14 = 0$
$\Rightarrow 2x - 10 = 0$
$\Rightarrow 2x = 10$
$\Rightarrow x = +5$
$H_4P_2O_5$: $4 + 2x - 10 = 0$
$\Rightarrow 2x - 6 = 0$
$\Rightarrow 2x = 6$
$\Rightarrow x = +3$
$H_4P_2O_6$: $\Rightarrow 4 + 2x - 12 = 0$
$\Rightarrow 2x - 8 = 0$
$\Rightarrow 2x = 8$
$\Rightarrow x = +4$

Q35: For a reaction of order n, the unit of the rate constant is:

(A) $mol^{1-n} L^{1-n} s$

(B) $mol^{1-n} L^{2n} s^{-1}$

(C) $mol^{1-n} L^{n-1} s^{-1}$

(D) $mol^{1-n} L^{1-n} s^{-1}$

Q35: For a reaction of order n, the unit of the rate constant is:

(A) $mol^{1-n} L^{1-n} s$

(B) $mol^{1-n} L^{2n} s^{-1}$

(C) $mol^{1-n} L^{n-1} s^{-1}$

(D) $mol^{1-n} L^{1-n} s^{-1}$

Solution:

The rate law can be written as Rate = k $[A]^n$

Using dimensional analysis –

$$\frac{\mathrm{mol}}{\mathrm{L\ .\ s}} = \mathrm{k}\frac{(\mathrm{mol})^{\mathrm{n}}}{(\mathrm{L})^{\mathrm{n}}}$$

$$\Rightarrow \mathrm{k} = \frac{\mathrm{mol}(\mathrm{L})^{\mathrm{n}}}{(\mathrm{mol})^{\mathrm{n}}\mathrm{L\ .\ s}} = \mathrm{mol}^{1\text{-}\mathrm{n}}\mathrm{L}^{\mathrm{n}\text{-}1}\mathrm{s}^{\text{-}1}$$

Q36: Given below are two statements:

Statement I: Aniline is less basic than acetamide.

Statement II: In aniline, the lone pair of electrons on nitrogen atom is delocalized over benzene ring due to resonance and hence less available to a proton.

Choose the **most appropriate** option :

(A) Statement I is true, but statement II is false.

(B) Statement I is false, but statement II is true.

(C) Both statement I and statement II are true.

(D) Both statement I and statement II are false.

Q36: Given below are two statements:

Statement I: Aniline is less basic than acetamide.

Statement II: In aniline, the lone pair of electrons on nitrogen atom is delocalized over benzene ring due to resonance and hence less available to a proton.

Choose the **most appropriate** option :

(A) Statement I is true, but statement II is false.

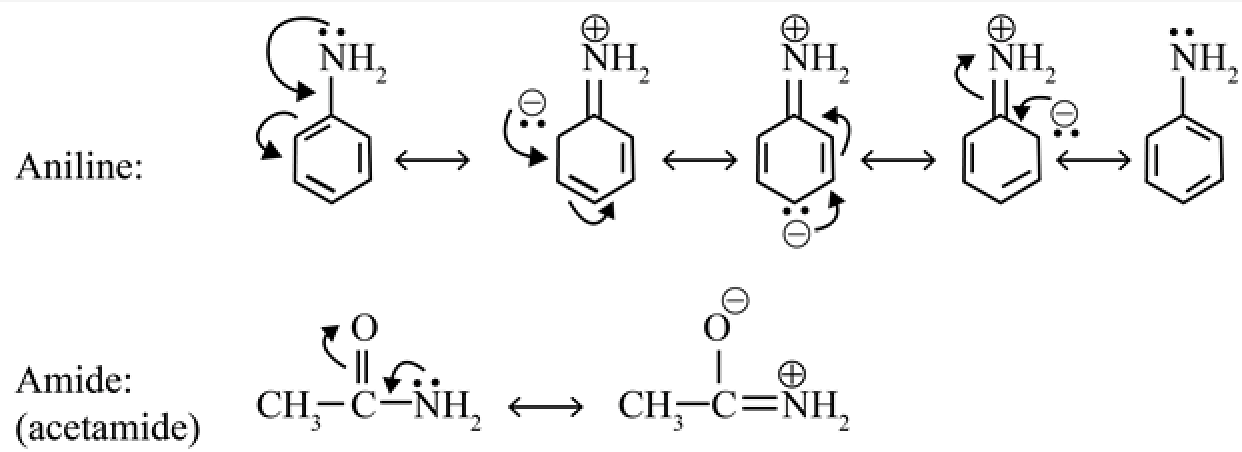
(B) Statement I is false, but statement II is true.

(C) Both statement I and statement II are true.

(D) Both statement I and statement II are false.

Solution:

Acetamide is less basic than aniline because its lone pair of Nitrogen atoms involved in resonance with oxygen atom hence it is less available to donate



Q37: The type of hybridization and magnetic property of the complex $[MnCl_6]^{3-}$, respectively, are:

(A) sp^3d^2 and diamagnetic

(B) d^2sp^3 and diamagnetic

(C) d^2sp^3 and paramagnetic

(D) sp^3d^2 and paramagnetic

Q37: The type of hybridization and magnetic property of the complex $[MnCl_6]^{3-}$, respectively, are:

(A) sp^3d^2 and diamagnetic

(B) d^2sp^3 and diamagnetic

(C) d^2sp^3 and paramagnetic

(D) sp^3d^2 and paramagnetic

Solution:

$[MnCl_6]^{3-} : Mn^{3+} = [Ar]\,3d^4$, in presence of weak field ligand, the electrons will not be paired against Hund's rule and hybridisation will be sp^3d^2 and there will be four unpaired electrons which makes the complex paramagnetic.

Q38: The number of geometrical isomers found in the metal complexes $[PtCl_2(NH_3)_2]$, and $[Ni(CO)_4]$, $[Ru(H_2O)_3Cl_3]$ and $[CoCl_2(NH_3)_4]^+$ respectively, are

(A) 1, 1, 1, 1

(B) 2, 1, 2, 2

(C) 2, 0, 2, 2

(D) 2, 1, 2, 1

Q38: The number of geometrical isomers found in the metal complexes $[PtCl_2(NH_3)_2]$, and $[Ni(CO)_4]$, $[Ru(H_2O)_3Cl_3]$ and $[CoCl_2(NH_3)_4]^+$ respectively, are

(A) 1, 1, 1, 1

(B) 2, 1, 2, 2

(C) 2, 0, 2, 2

(D) 2, 1, 2, 1

Solution:

$[PtCl_2(NH_3)_2]$ – square planar – cis and trans isomers
$[Ni(CO)_4]$ – tetrahedral – no geometrical isomer.
$[Ru(H_2O)_3Cl_3]$ – octahedral – fac and mer isomers
$[CoCl_2(NH_3)_4]^+$ – octahedral – cis and trans isomers

Q39: Which one of the following statements is **NOT** correct?

(A) Eutrophication indicates that water body is polluted

(B) The dissolved oxygen concentration below 6 ppm inhibits fish growth

(C) Eutrophication leads to increase in the oxygen level in water

(D) Eutrophication leads to anaerobic conditions

Q39: Which one of the following statements is **NOT** correct?

(A) Eutrophication indicates that water body is polluted

(B) The dissolved oxygen concentration below 6 ppm inhibits fish growth

(C) Eutrophication leads to increase in the oxygen level in water

(D) Eutrophication leads to anaerobic conditions

Solution:

Eutrophication is the process in which a water body becomes overly enriched with nutrients, leading to plentiful growth of simple plant life. The excessive growth (or bloom) of algae and plankton in a water body are indicators of this process. The excessive growth of algae in entrap, water is accompanied by the generation of a lame biomass of dead algae. These dead algae sink to the bottom of the water body where they are broken down by bacteria, which consume oxygen in the process.

Q40: Given below are two statements:

Statement I: Rutherford's gold foil experiment cannot explain the line spectrum of hydrogen atom.

Statement II: Bohr’s model of hydrogen atom contradicts Heisenberg’s uncertainty principles.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(A) **Statement I** is false, but **statement II** is true.

(B) **Statement I** is true, but **statement II** is false.

(C) Both **Statement I** and **statement II** are false

(D) Both **Statement I** and **statement II** are true

Q40: Given below are two statements:

Statement I: Rutherford's gold foil experiment cannot explain the line spectrum of hydrogen atom.

Statement II: Bohr’s model of hydrogen atom contradicts Heisenberg’s uncertainty principles.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(A) **Statement I** is false, but **statement II** is true.

(B) **Statement I** is true, but **statement II** is false.

(C) Both **Statement I** and **statement II** are false

(D) Both **Statement I** and **statement II** are true

Solution:

Rutherford's gold foil experiment cannot explain the line spectrum of hydrogen atom as it does not consider stationery orbits. Bohr’s model of hydrogen atom contradicts Heisenberg’s uncertainty principles as it tells us exact position and velocity of electron at the same time which is not possible according to Heisenberg’s principle.

Q41: Presence of which reagent will affect the reversibility of the following reaction, and change it to an irreversible reaction:

$$CH_4 + I_2 \underset{\text{Reversible}}{\overset{h\nu}{\rightleftharpoons}} CH_3-I + HI$$

(A) HOCl

(B) dilute HNO_2

(C) Liquid NH_3

(D) Concentrated HIO_3

Q41: Presence of which reagent will affect the reversibility of the following reaction, and change it to an irreversible reaction:

$$CH_4 + I_2 \underset{\text{Reversible}}{\overset{h\nu}{\rightleftharpoons}} CH_3-I + HI$$

(A) HOCl

(B) dilute HNO_2

(C) Liquid NH_3

(D) Concentrated HIO_3

Solution:

$5HI + \text{Conc. } HIO_3 \rightarrow 3I_2 + 3H_2O$

Q42: Which one among the following chemical tests is used to distinguish monosaccharide from disaccharide?

(A) Seliwanoff's test

(B) Iodine test

(C) Barfoed test

(D) Tollen's test

Q42: Which one among the following chemical tests is used to distinguish monosaccharide from disaccharide?

(A) Seliwanoff's test

(B) Iodine test

(C) Barfoed test

(D) Tollen's test

Solution:

The ‘Barfoed's test’ is utilized to separate lessening monosaccharide from a disaccharide sugar. The response is directed in a marginally acidic medium. A combination of ethanoic (acidic) corrosive and copper acetic acid derivation, is added to the test arrangement and bubbled.

Q43: Match **List-I** with **List-II.**

List – I(Drug)	**List – II(Class of Drug)**
(a) Furacin	(i) Antibiotic
(b) Arsphenamine	(ii) Tranquilizers
(c) Dimetone	(iii) Antiseptic
(d) Valium	(iv) Synthetic antihistamines

Choose the **most appropriate** answer from the options given below:

(A) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)

(B) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

(C) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

(D) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

Q43: Match **List-I** with **List-II.**

List – I(Drug)	**List – II(Class of Drug)**
(a) Furacin	(i) Antibiotic
(b) Arsphenamine	(ii) Tranquilizers
(c) Dimetone	(iii) Antiseptic
(d) Valium	(iv) Synthetic antihistamines

Choose the **most appropriate** answer from the options given below:

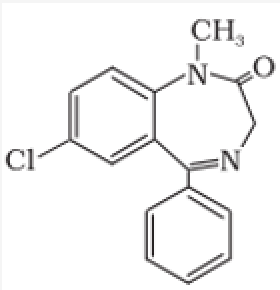
(A) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)

(B) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

(C) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

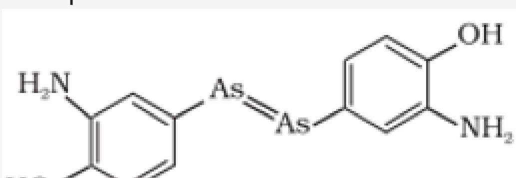
(D) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

Solution:



Valium

Tranquilizer



Salvarsan

Arsphenamine, known as a salvarsan used as Antibiotic.

Antiseptics are applied to the living tissues such as wounds, cuts, ulcers and diseased skin surfaces. Examples are furacine, formicine.

Q44: The statement that is INCORRECT about Ellingham diagram is

(A) provides idea about the reaction rate.

(B) provides idea about free energy change

(C) provides idea about changes in the phases during the reaction

(D) provides idea about reduction of metal oxide.

Q44: The statement that is INCORRECT about Ellingham diagram is

(A) provides idea about the reaction rate.

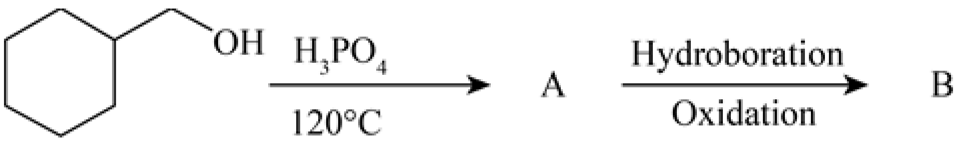
(B) provides idea about free energy change

(C) provides idea about changes in the phases during the reaction

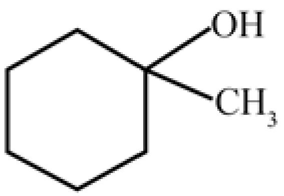
(D) provides idea about reduction of metal oxide.

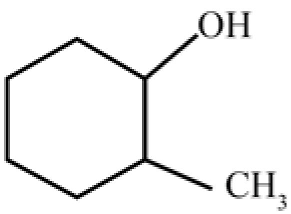
Solution:

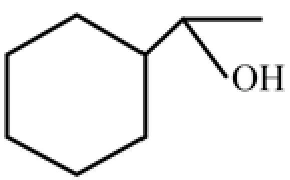
Ellingham diagram does not give any information about rate of reaction

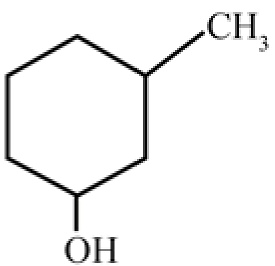
Q45: 

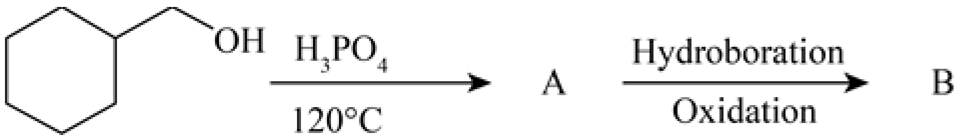
Consider the above reaction and identify the Product B:

(A) 

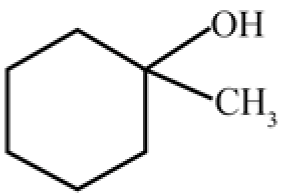
(B) 

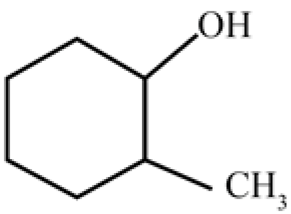
(C) 

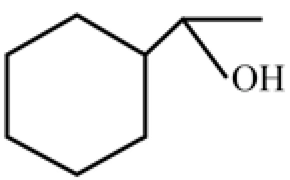
(D) 

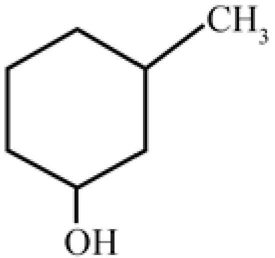
Q45: 

Consider the above reaction and identify the Product B:

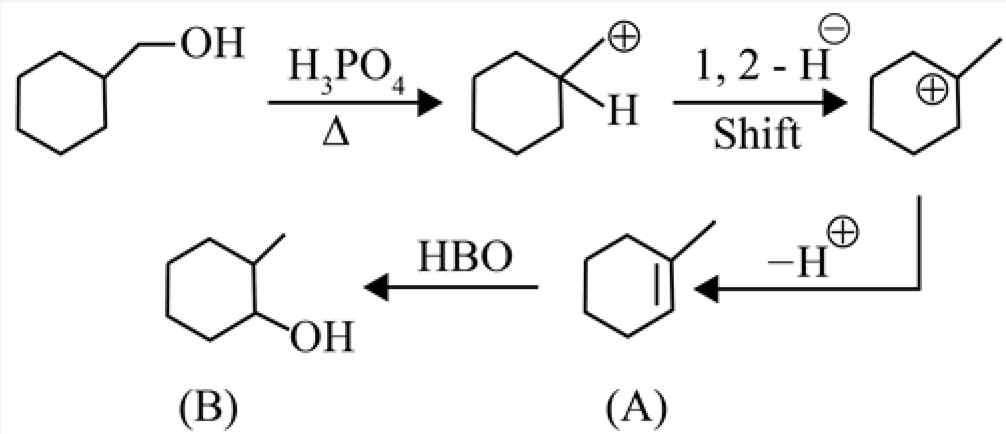
(A) 

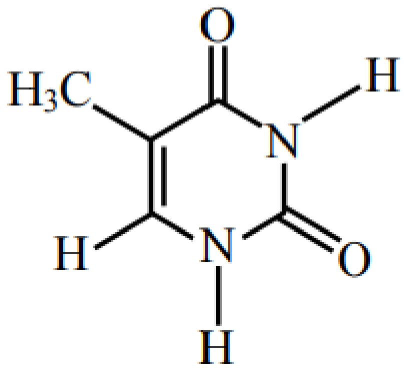
(B) 

(C) 

(D) 

Solution:



Q46: 

(A)

The compound 'A' is a complementary base of __________ in DNA strands.

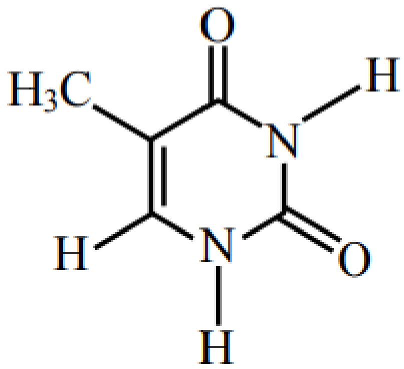
(A) Uracil

(B) Guanine

(C) Adenine

(D) Cytosine

Q46:



(A)

The compound 'A' is a complementary base of __________ in DNA strands.

(A) Uracil

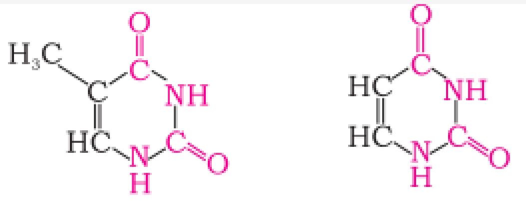
(B) Guanine

(C) Adenine

(D) Cytosine

Solution:

DNA contains four bases viz. adenine (A), guanine (G), cytosine (C) and thymine (T). RNA also contains four bases, the first three bases are same as in DNA but the fourth one is uracil (U).



Thymine (T) Uracil (U)

Q47: Staggered and eclipsed conformers of ethane are:

(A) Polymers

(B) Rotamers

(C) Enantiomers

(D) Mirror images

Q47: Staggered and eclipsed conformers of ethane are:

(A) Polymers

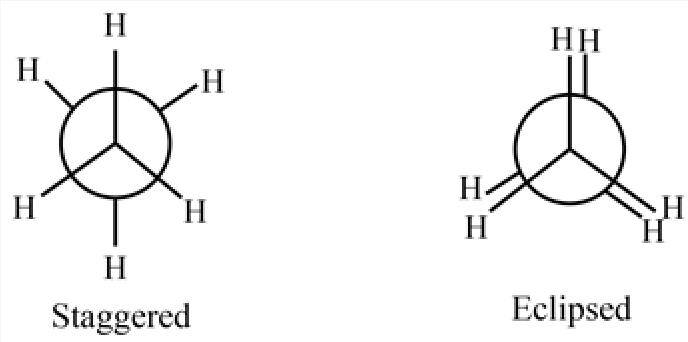
(B) Rotamers

(C) Enantiomers

(D) Mirror images

Solution:

Rotamers are any of a number of isomers of a molecule which can be inter converted by rotation of part of the molecule about a particular bond.



Staggered and Eclipsed form of ethane are inter convertible by rotation of the molecule about a bond. Hence these are rotamers.

Q48: Match **List-I** with **List-II**:

List – I	**List – II**
(a) NaOH	(i) Acidic
(b) $Be(OH)_2$	(ii) Basic
(c) $Ca(OH)_2$	(iii) Amphoteric
(d) $B(OH)_3$	
(e) $Al(OH)_3$	

Choose the **most appropriate** answer from the options given below:

(A) (a)-(ii), (b)-(ii), (c)-(iii), (d)-(ii), (e)-(iii)

(B) (a)-(ii), (b)-(iii), (c)-(ii), (d)-(i), (e)-(iii)

(C) (a)-(ii), (b)-(ii), (c)-(iii), (d)-(i), (e)-(iii)

(D) (a)-(ii), (b)-(i), (c)-(ii), (d)-(iii), (e)-(iii) .

Q48: Match **List-I** with **List-II**:

List – I	**List – II**
(a) NaOH	(i) Acidic
(b) $Be(OH)_2$	(ii) Basic
(c) $Ca(OH)_2$	(iii) Amphoteric
(d) $B(OH)_3$	
(e) $Al(OH)_3$	

Choose the **most appropriate** answer from the options given below:

(A) (a)-(ii), (b)-(ii), (c)-(iii), (d)-(ii), (e)-(iii)

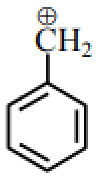
(B) (a)-(ii), (b)-(iii), (c)-(ii), (d)-(i), (e)-(iii)

(C) (a)-(ii), (b)-(ii), (c)-(iii), (d)-(i), (e)-(iii)

(D) (a)-(ii), (b)-(i), (c)-(ii), (d)-(iii), (e)-(iii) .

Solution:

NaOH $\Rightarrow$ Basic
$Be(OH)_2 \Rightarrow$ Amphoteric
$Ca(OH)_2 \Rightarrow$ Basic
$B(OH)_3 \Rightarrow$ Acidic
$Al(OH)_3 \Rightarrow$ Amphoteric

Q49:  $CH_2=\overset{\oplus}{C}H$ $CH_3-\overset{\oplus}{C}H_2$ $HC\equiv\overset{\oplus}{C}$

A B C D

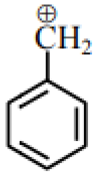
The correct order of stability of given carbocation is:

(A) A > C > B > D

(B) D > B > C > A

(C) D > B > A > C

(D) C > A > D > B

Q49:  $CH_2{=}\overset{\oplus}{C}H$ $CH_3{-}\overset{\oplus}{C}H_2$ $HC{\equiv}\overset{\oplus}{C}$

A B C D

The correct order of stability of given carbocation is:

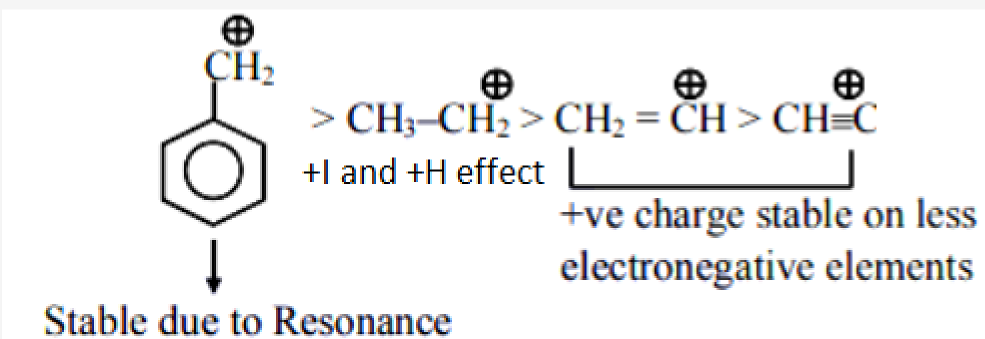
(A) A > C > B > D

(B) D > B > C > A

(C) D > B > A > C

(D) C > A > D > B

Solution:



Q50: Given below are two statements: One is labelled as **Assertion A** and the other labelled as **Reason R.**

Assertion A: Lithium halides are somewhat covalent in nature.

Reason R: Lithium possess high polarization capability.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) **A** is true but **R** is false

(B) **A** is false but **R** is true

(C) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**

(D) Both **A** and **R** are true and **R** is the correct explanation of **A**

Q50: Given below are two statements: One is labelled as **Assertion A** and the other labelled as **Reason R.**

Assertion A: Lithium halides are somewhat covalent in nature.

Reason R: Lithium possess high polarization capability.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) **A** is true but **R** is false

(B) **A** is false but **R** is true

(C) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**

(D) Both **A** and **R** are true and **R** is the correct explanation of **A**

Solution:

Lithium halides are somewhat covalent. It is because of the high polarisation capability of lithium ion (The distortion of electron cloud of the anion by the cation is called polarisation). The Li^+ ion is very small in size and has high tendency to distort electron cloud around the negative halide ion.

Q51: The density of NaOH solution is 1.2 g cm^{-3}. The molality of this solution is __________ m: (Round off to the Nearest Integer)

(Use: Atomic masses Na: 23.0 u, O: 16.0 u, H: 1.0 u, Density of H_2O: 1.0 g cm^{-3})

Q51: The density of NaOH solution is 1.2 g cm^{-3}. The molality of this solution is __________ m: (Round off to the Nearest Integer)
(Use: Atomic masses Na: 23.0 u, O: 16.0 u, H: 1.0 u, Density of H_2O: 1.0 g cm^{-3})

5.00

Solution:

Let the volume of solution = 1000 cm^3
∴ Weight of solution = 1200 g
Now, weight of water in the solution = 1000 g
Thus, weight of NaOH = 200 g
∴ Moles of NaOH = $\frac{200}{40}$ = 5
Thus, molarity = $\frac{5}{1}$ = 5 M

Q52: CO_2 gas adsorbs on charcoal following Freundlich adsorption isotherm. For a given amount of charcoal, the mass of CO_2 adsorbed becomes 64 times when the pressure CO_2 is doubled. The value of n in the Freundlich isotherm equation is ___________ $\times 10^{-2}$. (Round off to the Nearest Integer)

Q52: CO_2 gas adsorbs on charcoal following Freundlich adsorption isotherm. For a given amount of charcoal, the mass of CO_2 adsorbed becomes 64 times when the pressure CO_2 is doubled. The value of n in the Freundlich isotherm equation is __________ $\times 10^{-2}$. (Round off to the Nearest Integer)

17.00

Solution:

According to Freundlich adsorption isotherm equation –

$\frac{x}{m} \propto (p)^{1/n}$

$\Rightarrow \frac{x_1}{x_2} = \left(\frac{p_1}{p_2}\right)^{1/n}$

$\Rightarrow \frac{1}{64} = \left(\frac{1}{2}\right)^{1/n}$

$\Rightarrow \left(\frac{1}{2}\right)^6 = \left(\frac{1}{2}\right)^{1/n}$

$\Rightarrow \frac{1}{n} = 6 \Rightarrow n = \frac{1}{6} = 16.67 \times 10^{-2}$

Q53: The conductivity of a weak acid HA of concentration 0.00 l mol L^{-1} is 2.0×10^{-5} S cm^{-1}. If Λ_m^0 (HA) = 190 S cm^2 mol^{-1}, the ionization constant (K_a) of HA is equal to ___ $\times 10^{-6}$. (Round off to the Nearest Integer)

Q53: The conductivity of a weak acid HA of concentration 0.00 I mol L^{-1} is 2.0 × 10^{-5} S cm^{-1}. If Λ_m^0 (HA) = 190 S cm^2 mol^{-1}, the ionization constant (K_a) of HA is equal to ____ × 10^{-6}. (Round off to the Nearest Integer)

12.00

Solution:

Given, $\kappa = 2 \times 10^{-5} S\ cm^{-1}$

Thus, $\Lambda_m = \frac{\mathrm{K} \times 1000}{M} = \frac{2 \times 10^{-5} \times 10^3}{10^{-3}} = 20$

$\alpha = \frac{\Lambda_m}{\Lambda_m{}^\circ} = \frac{20}{190} = \frac{2}{19}$

Now, $K_a = \frac{C\alpha^2}{1-\alpha} = \frac{10^{-3} \times \frac{4}{361}}{\frac{17}{19}} = \frac{4}{19 \times 17} \times 10^{-3}$

$K_a = 0.01238 \times 10^{-3} = 12.38 \times 10^{-6}$

Q54: 1.46 g of a biopolymer dissolved in a 100 mL water at 300 K exerted an osmotic pressure of 2.42×10^{-3} bar. The molar mass of the biopolymer is _______ $\times 10^{4}$ g mol^{-1} (Round off to the Nearest Integer)

[Use: R = 0.083 L bar mol^{-1} K^{-1}]

Q54: 1.46 g of a biopolymer dissolved in a 100 mL water at 300 K exerted an osmotic pressure of 2.42×10^{-3} bar. The molar mass of the biopolymer is _______ $\times 10^4$ g mol^{-1} (Round off to the Nearest Integer)
[Use: R = 0.083 L bar mol^{-1} K^{-1}]

15.00

Solution:

$$\pi = \frac{wRT}{MV}$$
$$M = \frac{wRT}{\pi V} = \frac{1.46 \times 0.083 \times 300}{2.42 \times 10^{-3} \times 0.1} = 15.02 \times 10^4 \text{ g mol}^{-1}$$

Q55: An organic compound is subjected to chlorination to get compound A using 5.0 g of chlorine. When 0.5 g of compound A is reacted with $AgNO_3$ [Carius method], the percentage of compound A is ________ when it forms0.3849 g of AgCl. (Round off to the nearest integer (Atomic masses of Ag and Cl are 107.87 and 35.5 respectively)

Q55: An organic compound is subjected to chlorination to get compound A using 5.0 g of chlorine. When 0.5 g of compound A is reacted with $AgNO_3$ [Carius method], the percentage of compound A is ________ when it forms0.3849 g of AgCl. (Round off to the nearest integer (Atomic masses of Ag and Cl are 107.87 and 35.5 respectively)

19.00

Solution:

Mass of organic compound = 0.5 gm.

Mass of formed AgCl = 0.3849 gm

$$\% \text{ of Cl } = \frac{\text{atomic mass of Cl} \times \text{mass formed AgCl}}{\text{molecular mass of AgCl} \times \text{mass of organic compound}} \times 100$$

= 19.06

= 19

Q56: The number of geometrical isomers possible in triamminetrinitrocobalt(III) is X and in trioxalatochromate(III) is Y. Then the value of X + Y is ________.

Q56: The number of geometrical isomers possible in triamminetrinitrocobalt(III) is X and in trioxalatochromate(III) is Y. Then the value of X + Y is ________.

2.00

Solution:

$[Co(NH_3)_3(NO_2)_3] \Rightarrow Ma_3b_3$
Total geometric isomers = 2
$[Cr(C_2O_4)_3]^{3-} \Rightarrow M(AA)_3$
Total geometric isomers = 0
Thus, total = 2
Please note that $Cr(C_2O_4)_3$ can exist in one form only and we do not consider it as geometric isomerism.

Q57: In gaseous triethylamine the “-C-N-C-“ bond angle is _________ degree.

Q57: In gaseous triethylamine the “-C-N-C-“ bond angle is _________ degree.

108.00

Solution:

In gaseous triethylamine nitrogen has 3 bonds pairs, 1 lone pair, so, by vsepr theory, it is sp^3 hybridised and bond angle is 108^0

Q58: For water at 100^{o}C and 1 bar, $\Delta_{\text{vap}}\text{H} - \Delta_{\text{vap}}\text{U} = \_\_\_\_\_\_\_\_ \times 10^{2}\,\text{J mol}^{-1}$ (Round off to the Nearest Integer) [Use : R = 8.31 J mol^{-1} K^{-1}]
[Assume volume of H_2O(l) is much smaller than volume of H_2O(g). Assume H_2O(g) treated as an ideal gas].

Q58: For water at 100^{o}C and 1 bar, $\Delta_{\mathrm{vap}}\mathrm{H} - \Delta_{\mathrm{vap}}\mathrm{U} =$ ________ $\times 10^2 \mathrm{J\ mol^{-1}}$ (Round off to the Nearest Integer) [Use : R = 8.31 J mol^{-1} K^{-1}]
[Assume volume of $H_2O(l)$ is much smaller than volume of $H_2O(g)$. Assume $H_2O(g)$ treated as an ideal gas].

31.00

Solution:

The equation corresponding to vapourisation of water is written as -

$H_2O\,(l) \rightarrow H_2O\,(g)$

Now, $\Delta_{\mathrm{vap}}\mathrm{H} - \Delta_{\mathrm{vap}}\mathrm{U} = \Delta \mathrm{n_g RT} = 1 \times 8.31 \times 373 = 3099.63 = 30.9963 \times 10^2\, Jmol^{-1}$

Q59: $\mathrm{PCl_5 \rightleftharpoons PCl_3 + Cl_2 ; K_c = 1.844}$

3.0 moles of PCl_5 is introduced in a 1 L vessel. The number of moles of PCl_5 at equilibrium is _______ $\times 10^{-3}$. (Round off to the Nearest Integer)

Q59: $PCl_5 \rightleftharpoons PCl_3 + \ Cl_2; K_c = 1.844$

3.0 moles of PCl_5 is introduced in a 1 L vessel. The number of moles of PCl_5 at equilibrium is ________ $\times 10^{-3}$. (Round off to the Nearest Integer)

1396.00

Solution:

	PCl_5	$\rightleftharpoons$	PCl_3	+	Cl_2
Initial	3		-		-
Eqm.	$3 - x$		x		x

Now $\frac{x^2}{3-x} = 1.844$

$\Rightarrow x^2 = 5.532 - 1.844x$

$\Rightarrow x^2 + 1.844x - 5.532 = 0$

$\Rightarrow x = \frac{-1.844 \pm \sqrt{(1.844)^2 + 4 \times 5.532}}{2}$

$\Rightarrow x = 1.6042$

Thus, moles of PCl_5 = 3 – x = 1.3958 = 1395.8×10^{-3}

Q60: The difference between bond orders of CO and $\mathrm{NO}^{\oplus}$ is $\frac{x}{2}$ where x = ________.

Q60: The difference between bond orders of CO and $\mathrm{NO}^{\oplus}$ is $\frac{x}{2}$ where x = ________.

0.00

Solution:

Both the species are isoelectronic and thus will have same molecular orbital configuration and the same bond order. Thus, x = 0.

Q61: If the area of the bounded region
$R=\left\{(x, y): \max \{0, \log_e x\} \leqslant y \leqslant 2^x, \frac{1}{2} \leqslant x \leqslant 2\right\}$
is $\alpha(\log_e 2)^{-1}+\beta(\log_e 2)+\gamma$, then the value of $(\alpha+\beta-2\gamma)^2$ is equal to :

(A) 4

(B) 1

(C) 8

(D) 2

Q61: If the area of the bounded region

$$R=\left\{(x,y):\max\{0,\ \log_e x\}\leqslant y\leqslant 2^x,\ \frac{1}{2}\leqslant x\leqslant 2\right\}$$

is $\alpha(\log_e 2)^{-1}+\beta(\log_e 2)+\gamma$, then the value of $(\alpha+\beta-2\gamma)^2$ is equal to :

(A) 4

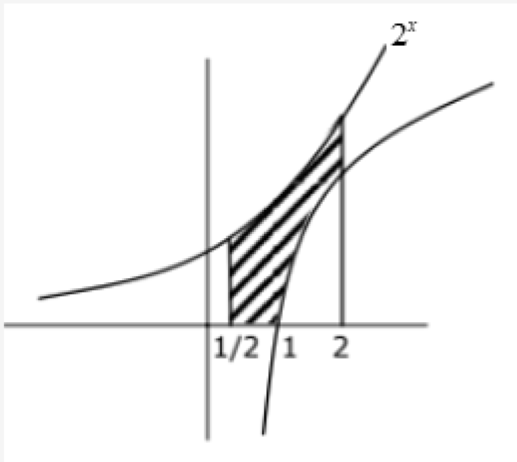
(B) 1

(C) 8

(D) 2

Solution:

$$R=\left\{(x,y):\max\{0,\ \log_e x\}\leqslant y\leqslant 2^x,\frac{1}{2}\leqslant x\leqslant 2\right\}$$



$$\int_{\frac{1}{2}}^{2} 2^x\,dx-\int_{1}^{2}\ell nx\,dx$$

$$=\left[\frac{2^x}{\ln 2}\right]_{\frac{1}{2}}^{2}-[x\ln x-x]_1^2$$

$$=\frac{(2^2)-2^{1/2}}{\log_e 2}-(2\ln 2-1)$$

$$\therefore \alpha=2^2-\sqrt{2},\beta=-2,\gamma=1$$

$$(\alpha+\beta-2\gamma)^2$$

$$=\left(2^2-\sqrt{2}-2-2\right)^2$$

$$=\left(\sqrt{2}\right)^2=2$$

Q62: Let α, β be two roots of the equation $x^2 + (20)^{\frac{1}{4}} x + (5)^{\frac{1}{2}} = 0$. Then $\alpha^8 + \beta^8$ is equal to:

(A) 10

(B) 50

(C) 160

(D) 100

Q62: Let α, β be two roots of the equation $x^2 + (20)^{\frac{1}{4}} x + (5)^{\frac{1}{2}} = 0$. Then $\alpha^8 + \beta^8$ is equal to:

(A) 10

(B) 50

(C) 160

(D) 100

Solution:

$$\left(x^2 + \sqrt{5}\right)^2 = \sqrt{20} x^2$$
$$x^4 = -5 \Rightarrow x^8 = 25$$
$$\alpha^8 + \beta^8 = 50$$

Q63: Let P and Q be two distinct points on a circle which has center at $C\left(2,3\right)$ and which passes through origin O, If OC is perpendicular to both the line segments CP and CQ, then the set $\{P, Q\}$ is equal to:

(A) $\{(-1,5),(5,1)\}$

(B) $\left\{\left(2+2\sqrt{2},3-\sqrt{5}\right),\left(2-2\sqrt{2},3+\sqrt{5}\right)\right\}$

(C) $\left\{\left(2+2\sqrt{2},3+\sqrt{5}\right),\left(2-2\sqrt{2},3-\sqrt{5}\right)\right\}$

(D) $\{(4,0),(0,6)\}$

Q63: Let P and Q be two distinct points on a circle which has center at $C(2,3)$ and which passes through origin O, If OC is perpendicular to both the line segments CP and CQ, then the set $\{P,Q\}$ is equal to:

(A) $\{(-1,5),(5,1)\}$

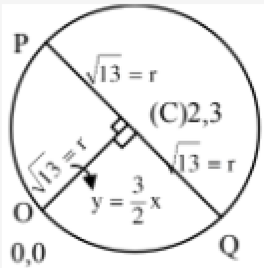
(B) $\{(2+2\sqrt{2},3-\sqrt{5}),(2-2\sqrt{2},3+\sqrt{5})\}$

(C) $\{(2+2\sqrt{2},3+\sqrt{5}),(2-2\sqrt{2},3-\sqrt{5})\}$

(D) $\{(4,0),(0,6)\}$

Solution:

Let θ be inclination of PQ



$\tan\theta = -\frac{2}{3}$

Using symmetric from of line]

$P, Q : \left(2 \pm \sqrt{13}\cos\theta, 3 \pm \sqrt{13}\sin\theta\right)$

$\left(2 \pm \sqrt{13}.\left(-\frac{3}{\sqrt{13}}\right), 3 \pm \sqrt{13}\left(\frac{2}{\sqrt{13}}\right)\right)$

$(-1, 5)\ \ \&\ \ (5, 1)$

Q64: Let $y = y(x)$ be solution of the differential equation $\log_e\left(\frac{dy}{dx}\right) = 3x + 4y$, with $y(0) = 0$. If $y\left(-\frac{2}{3}\log_e 2\right) = \alpha \log_e 2$, then the value of α is equal to :

(A) $-\frac{1}{2}$

(B) $-\frac{1}{4}$

(C) 2

(D) $\frac{1}{4}$

Q64: Let $y = y(x)$ be solution of the differential equation $\log_e\left(\frac{dy}{dx}\right) = 3x + 4y$, with $y(0) = 0$. If $y\left(-\frac{2}{3}\log_e 2\right) = \alpha \log_e 2$, then the value of α is equal to :

(A) $-\frac{1}{2}$

(B) $-\frac{1}{4}$

(C) 2

(D) $\frac{1}{4}$

Solution:

$\frac{dy}{dx} = e^{3x}.e^{4y} \Rightarrow \int e^{-4y}dy = \int e^{3x}dx$

$\frac{e^{-4y}}{-4} = \frac{e^{3x}}{3} + C$

Given, $y(0) = 0$

$\Rightarrow -\frac{1}{4} - \frac{1}{3} = C \Rightarrow C = -\frac{7}{12}$

$\frac{e^{-4y}}{-4} = \frac{e^{3x}}{3} - \frac{7}{12} \Rightarrow e^{-4y} = \frac{4e^{3x}-7}{-3}$

$e^{4y} = \frac{3}{7-4e^{3x}} \Rightarrow 4y = \ell n\left(\frac{3}{7-4e^{3x}}\right)$

$4y = \ell n\left(\frac{3}{6}\right)$ when $x = -\frac{2}{3}\ell n2$

$y = \frac{1}{4}\ell n\left(\frac{1}{2}\right) = -\frac{1}{4}\ell n2$

Q65: The compound statement $(P \vee Q) \wedge (\sim P) \Rightarrow Q$ is equivalent to:

(A) $P \vee Q$

(B) $\sim (P \Rightarrow Q) \Leftrightarrow P \wedge \sim Q$

(C) $P \wedge \sim Q$

(D) $\sim (P \Rightarrow Q)$

Q65: The compound statement $(P \vee Q) \wedge (\sim P) \Rightarrow Q$ is equivalent to:

(A) $P \vee Q$

(B) $\sim (P \Rightarrow Q) \Leftrightarrow P \wedge \sim Q$

(C) $P \wedge \sim Q$

(D) $\sim (P \Rightarrow Q)$

Solution:

Using Truth Table

P	Q	$P \vee Q$	$\sim P$	$(P \vee Q) \wedge \sim P$	$(P \vee Q) \wedge \sim P \rightarrow Q$	$\sim Q$	$P \wedge \sim Q$	$P \rightarrow Q$	$\sim (P \rightarrow Q)$
T	T	T	F	F	T	F	F	T	F
T	F	T	F	F	T	T	T	F	T
F	T	T	T	T	T	F	F	T	F
F	F	F	T	F	T	T	F	T	F

$P \wedge \sim Q$	$\sim (P \rightarrow Q) \Leftrightarrow P \wedge \sim Q$
F	T
T	T
F	T
F	T

Q66: Let
$A=\{(x,y)\in R\times R|2x^2+2y^2-2x-2y=1\}$,
$B=\{(x,y)\in R\times R|4x^2+4y^2-16y+7=0\}$ and
$C=\{(x,y)\in R\times R|x^2+y^2-4x-2y+5\leqslant r^2\}$.
Then the minimum value of $|r|$ such that $A\cup B\subseteq C$ is equal to:

(A) $\frac{3+\sqrt{10}}{2}$

(B) $1+\sqrt{5}$

(C) $\frac{2+\sqrt{10}}{2}$

(D) $\frac{3+2\sqrt{5}}{2}$

Q66: Let
$A=\{(x,y)\in R\times R|2x^2+2y^2-2x-2y=1\}$,
$B=\{(x,y)\in R\times R|4x^2+4y^2-16y+7=0\}$ and
$C=\{(x,y)\in R\times R|x^2+y^2-4x-2y+5\leqslant r^2\}$.
Then the minimum value of $|r|$ such that $A\cup B\subseteq C$ is equal to:

(A) $\frac{3+\sqrt{10}}{2}$

(B) $1+\sqrt{5}$

(C) $\frac{2+\sqrt{10}}{2}$

(D) $\frac{3+2\sqrt{5}}{2}$

Solution:

$S_1 : x^2 + y^2 - x - y - \frac{1}{2} = 0; C_1\left(\frac{1}{2}, \frac{1}{2}\right)$

$r_1 = \sqrt{\frac{1}{4} + \frac{1}{4} + \frac{1}{2}} = 1$

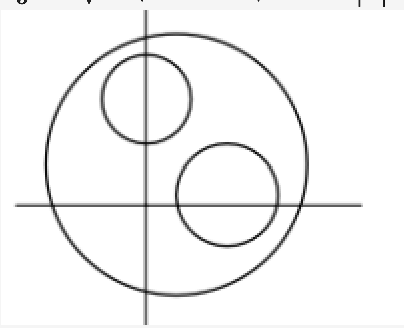
$S_2 : x^2 + y^2 - 4y + \frac{7}{4} = 0; C_2 : (0, 2)$

$r_2 \sqrt{4 - \frac{7}{4}} = \frac{3}{2}$

$S_3 : x^2 + y^2 - 4x - 2y + 5 - r^2 = 0$

$C_3 : (2, 1)$

$r_3 = \sqrt{4 + 1 - 5 + r^2} = |r|$



$C_1C_3 = \sqrt{\frac{5}{2}}$

$$\sqrt{\frac{5}{2}} \leqslant |r - 1| \Rightarrow \left.\begin{matrix} r \leqslant 1 + \sqrt{\frac{5}{2}} \\ r \geqslant \frac{3}{2} + \sqrt{5} \end{matrix}\right\} \qquad \ldots.(1)$$

$C_2C_3 = \sqrt{5} \leqslant \left|r - \frac{3}{2}\right|$

$$\left.\begin{matrix} r - \frac{3}{2} \geqslant \sqrt{5} \\ r - \frac{3}{2} \leqslant -\sqrt{5} \end{matrix}\right\} \qquad \ldots.(2)$$

From (1) & (2)

$$r_{\min} = \frac{3+2\sqrt{5}}{2}$$

Q67: The probability that a randomly selected 2 digit number belongs to the set $(n \in N : (2^n - 2)$ is a multiple of $3)$is equal to :

(A) $\frac{1}{2}$

(B) $\frac{1}{3}$

(C) $\frac{2}{3}$

(D) $\frac{1}{6}$

Q67: The probability that a randomly selected 2 digit number belongs to the set $(n \in N : (2^n - 2)$ is a multiple of $3)$is equal to :

(A) $\frac{1}{2}$

(B) $\frac{1}{3}$

(C) $\frac{2}{3}$

(D) $\frac{1}{6}$

Solution:

Total number of cases $= {}^{90}C_1 = 90$
Now $2^n - 2 = (3-1)^n - 2$
$= {}^nC_0 3^n - {}^nC_1 .3^{n-1} + \ldots + (-1)^{n-1}.\ {}^nC_{n-1} 3 + (-1)^n . {}^nC_n - 2$
$= 3\left(3^{n-1} - n3^{n-2} + \ldots + (-1)^{n-1}.n\right) + (-1)^n - 2$
$(2^n - 2)$ is multiply of 3 only when n is odd
Req. Probability $= \frac{45}{90} = \frac{1}{2}$

Q68: Let $A = \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix}$. If $A^{-1} = \alpha I + \beta A,\ \alpha, \beta \in R, I$ is a 2×2 identity matrix, then $4(\alpha - \beta)$ is equal to:

(A) 5

(B) 4

(C) 2

(D) $\frac{8}{3}$

Q68: Let $A = \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix}$. If $A^{-1} = \alpha I + \beta A,\ \alpha, \beta \in R, I$ is a 2×2 identity matrix, then $4(\alpha - \beta)$ is equal to:

(A) 5

(B) 4

(C) 2

(D) $\frac{8}{3}$

Solution:

$$A=\begin{bmatrix}1 & 2\\ -1 & 4\end{bmatrix}, |A|=6$$

$$A^{-1}=\frac{adjA}{|A|}=\frac{1}{6}\begin{bmatrix}4 & -2\\ 1 & 1\end{bmatrix}=\begin{bmatrix}\frac{2}{3} & -\frac{1}{3}\\ \frac{1}{6} & \frac{1}{6}\end{bmatrix}$$

Given $A^{-1}=\alpha I+\beta A$

$$\Rightarrow\begin{bmatrix}\frac{2}{3} & -\frac{1}{3}\\ \frac{1}{6} & \frac{1}{6}\end{bmatrix}=\begin{bmatrix}\alpha & 0\\ 0 & \alpha\end{bmatrix}+\begin{bmatrix}\beta & 2\beta\\ -\beta & 4\beta\end{bmatrix}$$

$$\left.\begin{array}{r}\alpha+\beta=\dfrac{2}{3}\\ \beta=-\dfrac{1}{6}\end{array}\right\}\Rightarrow\alpha=\frac{2}{3}+\frac{1}{6}=\frac{5}{6}$$

$$4(\alpha-\beta)=4(1)=4$$

Q69: If the mean and variance of the following data:

6, 10, 7, 13, a, 12, b, 12 are 9 and $\frac{37}{4}$ respectively, then $(a-b)^2$ is equal to:

(A) 12

(B) 24

(C) 16

(D) 32

Q69: If the mean and variance of the following data:
6, 10, 7, 13, a, 12, b, 12 are 9 and $\frac{37}{4}$ respectively, then $(a-b)^2$ is equal to:

(A) 12

(B) 24

(C) 16

(D) 32

Solution:

Mean $= \frac{6+10+7+13+a+12+b+12}{8} = 9$

$60 + a + b = 72$

$a + b = 12 \quad \ldots(1)$

variance $= \frac{\sum {x_i}^2}{n} - \left(\frac{\sum x_i}{n}\right)^2 = \frac{37}{4}$

$\Sigma x_i^2 = 6^2 + 10^2 + 7^2 + 13^2 + a^2 + b^2 + 12^2 + 12^2$

$= a^2 + b^2 + 642$

$\frac{a^2+b^2+642}{8} - (9)^2 = \frac{37}{4}$

$\frac{a^2+b^2}{8} + \frac{321}{4} - 81 = \frac{37}{4}$

$\frac{a^2+b^2}{8} = 81 + \frac{37}{4} - \frac{321}{4}$

$\frac{a^2+b^2}{8} = 81 - 71$

$\therefore a^2 + b^2 = 80 \quad \ldots(2)$

From (1) $a^2 + b^2 + 2ab = 144$

$80 + 2ab = 144 \quad \therefore 2ab = 64$

$(a-b)^2 = a^2 + b^2 - 2ab = 80 - 64 = 16$

Q70: A ray of light through (2,1) is reflected at a point P on the y – axis and then passes through the point (5,3). If this reflected ray is the directrix of an ellipse with eccentricity $\frac{1}{3}$ and the distance of the nearer focus from this directrix is $\frac{8}{\sqrt{53}}$, then the equation of the other directrix can be :

(A) $2x - 7y - 39 = 0$ or $2x - 7y - 7 = 0$

(B) $11x + 7y + 8 = 0$ or $11x + y - 15 = 0$

(C) $2x - 7y + 29 = 0$ or $2x - 7y - 7 = 0$

(D) $11x - 7y - 8 = 0$ or $11x + 7y + 15 = 0$

Q70: A ray of light through (2,1) is reflected at a point P on the y – axis and then passes through the point (5,3). If this reflected ray is the directrix of an ellipse with eccentricity $\frac{1}{3}$ and the distance of the nearer focus from this directrix is $\frac{8}{\sqrt{53}}$, then the equation of the other directrix can be :

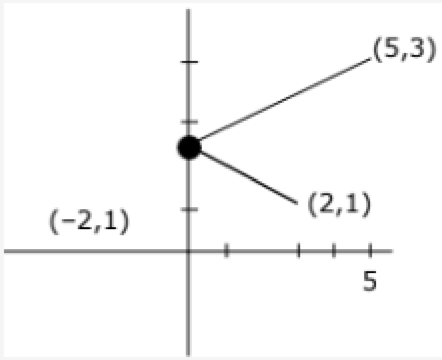
(A) $2x-7y-39=0$ or $2x-7y-7=0$

(B) $11x+7y+8=0$ or $11x+y-15=0$

(C) $2x-7y+29=0$ or $2x-7y-7=0$

(D) $11x-7y-8=0$ or $11x+7y+15=0$

Solution:



Equation of reflected Ray

$y-1=\frac{2}{7}(x+2)$

$7y-7=2x+4$

$2x-7y+11=0$

Let the equation of other directrix is

$2x-7y+\lambda=0$

Distance of directrix from focus $=\frac{8}{\sqrt{53}}$

$\frac{a}{e}-ae=\frac{8}{\sqrt{53}}$

$3a-\frac{a}{3}=\frac{8}{\sqrt{53}}$ or $a=\frac{3}{\sqrt{53}}$

Distance between two directrices $=\frac{2a}{e}$

$=2\times3\times\frac{3}{\sqrt{53}}=\frac{18}{\sqrt{53}}$

$\left|\frac{\lambda-11}{\sqrt{53}}\right|=\frac{18}{\sqrt{53}}$

$\lambda-11=18$ or -18

$\lambda=29$ or -7

Q71: Let $f:\left(-\frac{\pi}{4},\frac{\pi}{4}\right)\rightarrow R$ be defined as

$$f\left(x\right)=\begin{cases}\left(1+\left|\sin x\right|\right)^{\frac{3a}{\left|\sin x\right|}}, & -\frac{\pi}{4}<x<0\\ b, & x=0\\ e^{\cot 4x/\cot 2x}, & 0<x<\frac{\pi}{4}\end{cases}$$

If $f$is continuous at $x=0$, then the value of $6a+b^{2}$ is equal to:

(A) e

(B) $1+e$

(C) $1-e$

(D) $e-1$

Q71: Let $f:\left(-\frac{\pi}{4}, \frac{\pi}{4}\right) \rightarrow R$ be defined as

$$f(x)=\begin{cases}(1+|\sin x|)^{\frac{3a}{|\sin x|}}, & -\frac{\pi}{4}<x<0 \\ b, & x=0 \\ e^{\cot 4x / \cot 2x}, & 0<x<\frac{\pi}{4}\end{cases}$$

If f is continuous at $x=0$, then the value of $6a+b^2$ is equal to:

(A) e

(B) $1+e$

(C) $1-e$

(D) $e-1$

Solution:

$$\lim_{x \rightarrow 0} f(x)=b$$

$$\lim_{x \rightarrow 0^+} e^{\frac{\cot 4x}{\cot 2x}}=e^{\frac{1}{2}}=b$$

$$\lim_{x \rightarrow 0^-}(1+|\sin x|)^{\frac{3a}{|\sin x|}}=e^{3a}=e^{\frac{1}{2}}$$

$$a=\frac{1}{6} \Rightarrow 6a=1$$

$$\left(6a+b^2\right)=(1+e)$$

Q72: If $\sin\theta + \cos\theta = \frac{1}{2}$, then $16\left(\sin(2\theta) + \cos(4\theta) + \sin(6\theta)\right)$ is equal to :

(A) 27

(B) –27

(C) –23

(D) 23

Q72: If $\sin\theta + \cos\theta = \frac{1}{2}$, then $16\left(\sin(2\theta) + \cos(4\theta) + \sin(6\theta)\right)$ is equal to :

(A) 27

(B) –27

(C) –23

(D) 23

Solution:

$16\left[2\sin 4\theta \cos 2\theta + \cos 4\theta\right]$
$16\left[4\sin 2\theta \cos^2 2\theta + 2\cos^2 2\theta - 1\right]$
Now :
$\sin\theta + \cos\theta = \frac{1}{2}$
$1 + \sin 2\theta = \frac{1}{4}$
$\sin 2\theta = -\frac{3}{4}$
$\cos^2 2\theta = 1 - \frac{9}{16} = \frac{7}{16}$
$16\left[4\left(-\frac{3}{4}\right)\left(\frac{7}{16}\right) + 2\left(\frac{7}{16}\right) - 1\right]$
$16\left[\frac{-7}{16} - 1\right] \Rightarrow -23$

Q73: Let C be the set of all complex numbers. Let
$S_1 = \left\{z \in C \mid |z - 3 - 2i|^2 = 8\right\}$
$S_2 = \{z \in C \mid \operatorname{Re}(z) \geqslant 5\}$and
$S_3 = \{z \in C \mid |z - \bar{z}| \geqslant 8\}$.
Then the number of elements in $S_1 \cap S_2 \cap S_3$ is equal to:

(A) 1

(B) 0

(C) Infinite

(D) 2

Q73: Let C be the set of all complex numbers. Let

$S_1=\left\{z \in C \| z-3-2 i\right|^2=8\right\}$

$S_2=\{z \in C \mid \operatorname{Re}(z) \geqslant 5\}$and

$S_3=\{z \in C\|z-\bar{z}\| \geqslant 8\}$.

Then the number of elements in $S_1 \cap S_2 \cap S_3$ is equal to:

(A) 1

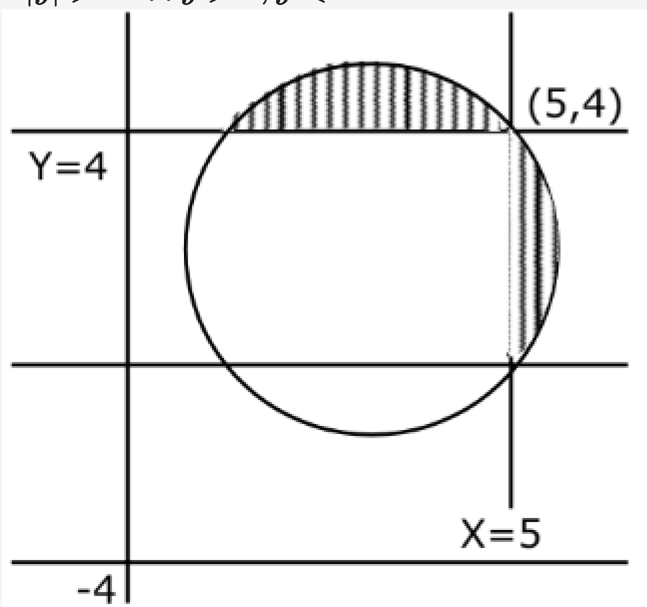
(B) 0

(C) Infinite

(D) 2

Solution:

$S_1 : |z-3-2i|^2 = 8$
$|z-3-2i| = 2\sqrt{2}$
$(x-3)^2 + (y-2)^2 = \left(2\sqrt{2}\right)^2$
$S_2 : x \geqslant 5$
$S_3 : |z-\bar{z}| \geqslant 8$
$|2iy| \geqslant 8$
$2|y| \geqslant 8 \quad \therefore y \geqslant 4, y \leqslant -4$



$n\left(S_1 \cap S_2 \cap S_3\right) = 1$

Q74: Let $\vec{a} = \hat{i} + \hat{j} + 2\hat{k}$ and $\vec{b} = -\hat{i} + 2\hat{j} + 3\hat{k}$. Then the vector product $\left(\vec{a} + \vec{b}\right) \times \left(\vec{a} \times \left(\left(\vec{a} - \vec{b}\right) \times \vec{b}\right)\right) \times \vec{b}$ is equal to:

(A) $5\left(30\hat{i} - 5\hat{j} + 7\hat{k}\right)$

(B) $5\left(34\hat{i} - 5\hat{j} + 3\hat{k}\right)$

(C) $7\left(30\hat{i} - 5\hat{j} + 7\hat{k}\right)$

(D) $7\left(34\hat{i} - 5\hat{j} + 3\hat{k}\right)$

Q74: Let $\vec{a} = \hat{i} + \hat{j} + 2\hat{k}$ and $\vec{b} = -\hat{i} + 2\hat{j} + 3\hat{k}$. Then the vector product $\left(\vec{a} + \vec{b}\right) \times \left(\vec{a} \times \left(\left(\vec{a} - \vec{b}\right) \times \vec{b}\right)\right) \times \vec{b}$ is equal to:

(A) $5\left(30\hat{i} - 5\hat{j} + 7\hat{k}\right)$

(B) $5\left(34\hat{i} - 5\hat{j} + 3\hat{k}\right)$

(C) $7\left(30\hat{i} - 5\hat{j} + 7\hat{k}\right)$

(D) $7\left(34\hat{i} - 5\hat{j} + 3\hat{k}\right)$

Solution:

$$\vec{a} = \hat{i} + \hat{j} + 2\hat{k}$$
$$\vec{b} = -\hat{i} + 2\hat{j} + 3\hat{k}$$
$$\vec{a} + \vec{b} = 3\hat{j} + 5\hat{k}; \vec{a} \cdot \vec{b} = -1 + 2 + 6 = 7$$
$$\left(\left(\vec{a} \times \left(\left(\vec{a} - \vec{b}\right) \times \vec{b}\right)\right) \times \vec{b}\right)$$
$$= \left(\left(\vec{a} \times \left(\vec{a} \times \vec{b} - \vec{b} \times \vec{b}\right)\right) \times \vec{b}\right)$$
$$= \left(\vec{a} \times \left(\vec{a} \times \vec{b} - 0\right)\right) \times \vec{b}$$
$$= \left(\vec{a} \times \left(\vec{a} \times \vec{b}\right)\right) \times \vec{b}$$
$$= \left(\left(\vec{a} \cdot \vec{b}\right)\vec{a} - (\vec{a} \cdot \vec{a})\vec{b}\right) \times \vec{b}$$
$$\left(\vec{a}.\vec{b}\right)\vec{a} \times \vec{b} - (\vec{a} \cdot \vec{a})\left(\vec{b} \times \vec{b}\right)$$
$$\left(\vec{a}.\vec{b}\right)\left(\vec{a} \times \vec{b}\right)$$
$$\vec{a} \times \vec{b} = \begin{vmatrix} i & j & k \\ 1 & 1 & 2 \\ -1 & 2 & 3 \end{vmatrix} = -\hat{i} - 5\hat{j} + 3\hat{k}$$
$$\therefore 7\left(-\hat{i} - 5\hat{j} + 3\hat{k}\right)$$
$$\left(\vec{a} + \vec{b}\right) \times \left(7\left(-\hat{i} - 5\hat{j} + 3\hat{k}\right)\right)$$
$$= 7\left(0\hat{i} + 3\hat{j} + 5\hat{k}\right) \times \left(-\hat{i} - 5\hat{j} + 3\hat{k}\right)$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 3 & 5 \\ -1 & -5 & 3 \end{vmatrix}$$

$$= 34\hat{i} - (5)\hat{j} + \left(3\hat{k}\right)$$

$$= 34\hat{i} - 5\hat{j} + 3\hat{k}$$

$$\therefore 7\left(34\hat{i} - 5\hat{j} + 3\hat{k}\right)$$

Q75: The value of the definite integral

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{dx}{\left(1+e^{x \cos x}\right)\left(\sin^4 x+\cos^4 x\right)}$$

is equal to :

(A) $\frac{\pi}{\sqrt{2}}$

(B) $-\frac{\pi}{4}$

(C) $\frac{\pi}{2\sqrt{2}}$

(D) $-\frac{\pi}{2}$

Q75: The value of the definite integral

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{dx}{\left(1+e^{x \cos x}\right)\left(\sin^4 x+\cos^4 x\right)}$$

is equal to :

(A) $\frac{\pi}{\sqrt{2}}$

(B) $-\frac{\pi}{4}$

(C) $\frac{\pi}{2\sqrt{2}}$

(D) $-\frac{\pi}{2}$

Solution:

$$I=\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\frac{dx}{\left(1+e^{x\cos x}\right)\left(\sin^4 x+\cos^4 x\right)}\ \ldots(1)$$

Using $\int_a^b f(x)\,dx=\int_a^b f(a+b-x)\,dx$

$$I=\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\frac{dx}{\left(1+e^{-\cos x}\right)\left(\sin^4 x+\cos^4 x\right)}$$

Add (1) and (2)

$$2I=\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\frac{dx}{\sin^4 x+\cos^4 x}$$

$$2I=2\int_{0}^{\frac{\pi}{4}}\frac{dx}{\sin^4 x+\cos^4 x}$$

$$I=\int_{0}^{\frac{\pi}{4}}\frac{\left(1+\tan^2 x\right)\sec^2 x}{\tan^4 x+1}dx$$

$$I=\int_{0}^{\frac{\pi}{4}}\frac{\left(1+\frac{1}{\tan^2 x}\right)\sec^2 x}{\left(\tan x-\frac{1}{\tan x}\right)^2+2}$$

$$\tan x-\frac{1}{\tan x}=t$$

$$\left(1+\frac{1}{\tan^2 x}\right)\sec^2 x dx=dt$$

$$I=\int_{-\infty}^{0}\frac{dt}{t^2+2}=\left[\frac{1}{2}\tan^{-1}\left(\frac{t}{\sqrt{2}}\right)\right]_{-\infty}^{0}$$

$$I=0-\frac{1}{\sqrt{2}}\left(-\frac{\pi}{2}\right)=\frac{\pi}{2\sqrt{2}}$$

Q76: Let the plane passing through the point (–1,0, –2) and perpendicular to each of the planes $2x + y - z = 2$ and $x - y - z = 3$ be $ax + by + cz + 8 = 0$, then the value of $a + b + c$ is equal to:

(A) 8

(B) 4

(C) 3

(D) 5

Q76: Let the plane passing through the point (–1,0, –2) and perpendicular to each of the planes $2x + y - z = 2$ and $x - y - z = 3$ be $ax + by + cz + 8 = 0$, then the value of $a + b + c$ is equal to:

(A) 8

(B) 4

(C) 3

(D) 5

Solution:

Normal of req. plane $\left(2\hat{i} + \hat{j} - \hat{k}\right) \times \left(\hat{i} - \hat{j} - \hat{k}\right) = -2\hat{i} + \hat{j} - 3\hat{k}$

Equation of plane

$-2(x+1) + 1(y-0) - 3(z+2) = 0$

$-2x + y - 3z - 8 = 0$

$2x - y + 3z + 8 = 0$

$a + b + c = 4$

Q77: The value of $\lim_{n\to\infty} \frac{1}{n} \sum_{j=1}^{n} \frac{(2j-1)+8n}{(2j-1)+4n}$ is equal to:

(A) $2 - \log_e\left(\frac{2}{3}\right)$

(B) $3 + 2\log_e\left(\frac{2}{3}\right)$

(C) $1 + 2\log_e\left(\frac{3}{2}\right)$

(D) $5 + \log_e\left(\frac{3}{2}\right)$

Q77: The value of $\lim\limits_{n\to\infty} \frac{1}{n} \sum\limits_{j=1}^{n} \frac{(2j-1)+8n}{(2j-1)+4n}$ is equal to:

(A) $2 - \log_e\left(\frac{2}{3}\right)$

(B) $3 + 2\log_e\left(\frac{2}{3}\right)$

(C) $1 + 2\log_e\left(\frac{3}{2}\right)$

(D) $5 + \log_e\left(\frac{3}{2}\right)$

Solution:

$$\lim_{n\to\infty} \frac{1}{n} \sum_{j=1}^{n} \frac{\left(\frac{2j}{n} - \frac{1}{n} + 8\right)}{\left(\frac{2j}{n} - \frac{1}{n} + 4\right)}$$

$$\int_0^1 \frac{2x+8}{2x+4} dx = \int_0^1 dx + \int_0^1 \frac{4}{2x+4} dx$$

$$= 1 + 4\tfrac{1}{2}(\ell n\,|2x+4|)_0^1$$

$$1 + 2\ell n\left(\frac{3}{2}\right)$$

Q78: Let $f : R \rightarrow R$ be a function such that $f(2) = 4$ and $f'(2) = 1$. Then, the value of $\lim_{x \rightarrow 2} \frac{x^2 f(2) - 4f(x)}{x-2}$ is equal to:

(A) 4

(B) 8

(C) 16

(D) 12

Q78: Let $f : R \rightarrow R$ be a function such that $f(2) = 4$ and $f'(2) = 1$. Then, the value of $\lim_{x \rightarrow 2} \frac{x^2 f(2) - 4f(x)}{x-2}$ is equal to:

(A) 4

(B) 8

(C) 16

(D) 12

Solution:

Apply L'Hopital Rule

$$\lim_{x \rightarrow 2} \left(\frac{2x f(2) - 4f'(x)}{1} \right)$$

$$= \frac{4(4) - 4}{1} = 12$$

Q79: If the coefficients of x^7 in $\left(x^2 + \frac{1}{bx}\right)^{11}$ and x^{-7} in $\left(x - \frac{1}{bx^2}\right)^{11}$, $b \neq 0$, are equal, then the value of b is equal to:

(A) –1

(B) 2

(C) –2

(D) 1

Q79: If the coefficients of x^7 in $\left(x^2 + \frac{1}{bx}\right)^{11}$ and x^{-7} in $\left(x - \frac{1}{bx^2}\right)^{11}$, $b \neq 0$, are equal, then the value of b is equal to:

(A) –1

(B) 2

(C) –2

(D) 1

Solution:

Coefficient of x^7 in $\left(x^2+\frac{1}{bx}\right)^{11}$

${}^{11}C_r\left(x^2\right)^{11-r}.\left(\frac{1}{bx}\right)^r$

${}^{11}C_r x^{22-3r}.\frac{1}{b^r}$

$22-3r=7$

$r=5$

$\therefore\ {}^{11}C_5.\frac{1}{b^5}.x^7$

Coefficient of x^{-7} in $\left(x-\frac{1}{bx^2}\right)^{11}$

${}^{11}C_r(x)^{11-r}.\left(-\frac{1}{bx^2}\right)^r$

${}^{11}C_r x^{11-3r}.\frac{(-1)^r}{b^r}$

$11-3r=-7\therefore r=6$

${}^{11}C_6.\frac{1}{b^6}x^{-7}$

${}^{11}C_5.\frac{1}{b^5}={}^{11}C_6.\frac{1}{b^6}$

Since $b\neq 0\ \therefore b=1$

Q80: Two tangents are drawn from the point $P(-1,1)$ to the circle $x^2+y^2-2x-6y+6=0$. If these tangents touch the circle at points A and B, and if D is a point on the circle such that length of the segments AB and AD are equal, then the area of the triangle ABD is eqaul to:

(A) 2

(B) $\left(3\sqrt{2}+2\right)$

(C) 4

(D) $3\left(\sqrt{2}-1\right)$

Q80: Two tangents are drawn from the point $P(-1, 1)$ to the circle $x^2 + y^2 - 2x - 6y + 6 = 0$. If these tangents touch the circle at points A and B, and if D is a point on the circle such that length of the segments AB and AD are equal, then the area of the triangle ABD is eqaul to:

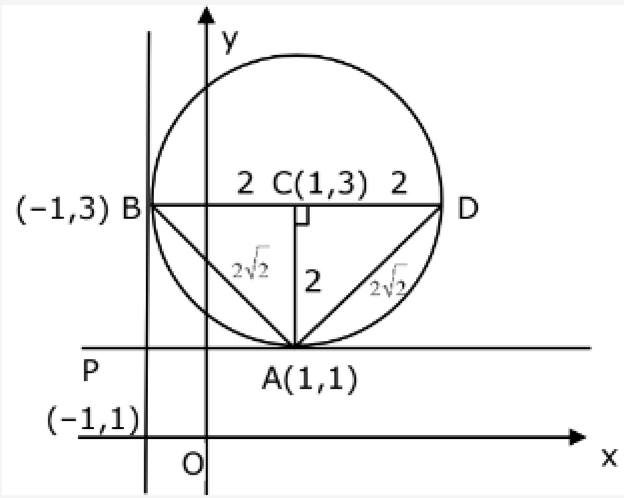
(A) 2

(B) $(3\sqrt{2} + 2)$

(C) 4

(D) $3(\sqrt{2} - 1)$

Solution:



$\Delta ABD = \frac{1}{2} \times 2 \times 4$

= 4

Q81: $f(x)=\begin{vmatrix} \sin^2 x & -2+\cos^2 x & \cos 2x \\ 2+\sin^2 x & \cos^2 x & \cos 2x \\ \sin^2 x & \cos^2 x & 1+\cos 2x \end{vmatrix}, x \in [0, \pi]$

Then the maximum value of $f(x)$ is equal to ______.

Q81: $f(x) = \begin{vmatrix} \sin^2 x & -2+\cos^2 x & \cos 2x \\ 2+\sin^2 x & \cos^2 x & \cos 2x \\ \sin^2 x & \cos^2 x & 1+\cos 2x \end{vmatrix}, x \in [0, \pi]$

Then the maximum value of $f(x)$ is equal to _____.

6.00

Solution:

$$\begin{vmatrix} -2 & -2 & 0 \\ 2 & 0 & -1 \\ \sin^2 x & \cos^2 x & 1+\cos 2x \end{vmatrix} \begin{pmatrix} R_1 \to R_1 - R_2 \\ \& \ R_2 \to R_2 - R_3 \end{pmatrix}$$

$$= -2\left(\cos^2 x\right) + 2\left(2 + 2\cos 2x + \sin^2 x\right)$$

$$= 4 + 4\cos 2x - 2\left(\cos^2 x - \sin^2 x\right)$$

$$f(x) = 4 + \underbrace{2\cos 2x}_{\max=1}$$

$$f(x)_{\max} = 4 + 2 = 6$$

Q82: Let $F:[3,5]\rightarrow R$ be a twice differentiable from function on $(3,5)$ such that
$F(x)=e^{-x}\int_3^x\left(3t^2+2t+4F'(t)\right)dt.$

If $F'(4)=\frac{\alpha e^{\beta}-224}{\left(e^{\beta}-4\right)^2}$, then $\alpha+\beta$ is equal to ______

Q82: Let $F:[3,5]\rightarrow R$ be a twice differentiable from function on $(3,5)$ such that
$F(x)=e^{-x}\int_3^x\left(3t^2+2t+4F'(t)\right)dt.$
If $F'(4)=\frac{\alpha e^{\beta}-224}{\left(e^{\beta}-4\right)^2}$, then $\alpha+\beta$ is equal to ______

16.00

Solution:

$$F(3)=0$$

$$e^{x}F(x)=\int_{3}^{x}\left(3t^{2}+2t+4F'(t)\right)dt$$

$$e^{x}F(x)+e^{x}F'(x)=3x^{2}+2x+4F'(x)$$

$$\left(e^{x}-4\right)\frac{dy}{dx}+e^{x}y=\left(3x^{2}+2x\right)$$

$$\frac{dy}{dx}+\frac{e^{x}y}{\left(e^{x}-4\right)}=\frac{\left(3x^{2}+2x\right)}{\left(e^{x}-4\right)}$$

$$ye^{\int\frac{e^{x}}{\left(e^{x}-4\right)}}=\int\left(\frac{3x^{2}+2x}{\left(e^{x}-4\right)}\right)e^{\int\frac{e^{x}}{\left(e^{x}-4\right)}dx}dx$$

$$y\left(e^{x}-4\right)=\int\left(3x^{2}+2x\right)dx+c$$

$$y\left(e^{x}-4\right)=x^{3}+x^{2}+c$$

Put $x=3\Rightarrow c=-36$

$$F(x)=\frac{\left(x^{3}+x^{2}-36\right)}{\left(e^{x}-4\right)}$$

$$F'(x)=\frac{\left(3x^{2}+2x\right)\left(e^{x}-4\right)-\left(x^{3}+x^{2}-36\right)e^{x}}{\left(e^{x}-4\right)^{2}}$$

$$F'(4)=\frac{56\left(e^{4}-4\right)-44e^{4}}{\left(e^{4}-4\right)^{2}}$$

$$=\frac{12e^{4}-224}{\left(e^{4}-4\right)^{2}}\Rightarrow\alpha=12$$

$$\beta=4$$

$$\alpha+\beta=16$$

Q83: Let a plane P pass through the point (3,7, –7) and contain the line, $\frac{x-2}{-3}=\frac{y-3}{2}=\frac{z+2}{1}$. If distance of the plane P from the origin is d, then d^2 is equal to

_______.

Q83: Let a plane P pass through the point (3,7, –7) and contain the line, $\frac{x-2}{-3}=\frac{y-3}{2}=\frac{z+2}{1}$. If distance of the plane P from the origin is d, then d^2 is equal to _______.

3.00

Solution:

Let $A \equiv (3,7,-7)$ and $B \equiv (2,3,-2)$

$\overrightarrow{BA}=\left(\hat{i}+4\hat{j}-5\hat{k}\right)$

$\vec{n}=-3\hat{i}+2\hat{j}+\hat{k}$

$\overrightarrow{BA}\times\vec{n}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ -3 & 2 & 1\\ 1 & 4 & -5\end{vmatrix}$

$a\hat{i}+b\hat{j}+c\hat{k}=-14\hat{i}-14\hat{j}-14\hat{k}$

$a:b:c=1:1:1$

Plane is $(x-3)+(y-7)+(z+7)=0$

$x+y+z-3=0$

$d=\sqrt{3}\Rightarrow d^2=3$

Q84: Let $S=\{1,2,3,4,5,6,7\}$ and N be the number of possible functions $f:S\rightarrow S$ such that $f(m\times n)=f(m)\times f(n)$ for every $m,\ n\in S$ and $m\times n\in S$. Find sum of digits of N.

Q84: Let $S = \{1, 2, 3, 4, 5, 6, 7\}$ and N be the number of possible functions $f : S \to S$ such that $f(m \times n) = f(m) \times f(n)$ for every $m,\ n \in S$ and $m \times n \in S$. Find sum of digits of N.

13.00

Solution:

$f(mn) = f(m) \,.\, f(n)$
Put $m = 1, f(n) = f(1) \,.\, f(n) \Rightarrow f(1) = 1$
Put $m = n = 2$

$$f(4) = f(2) \,.\, f(2) \begin{cases} f(2) = 1 \Rightarrow f(4) = 1 \\ \quad or \\ f(2) = 2 \Rightarrow f(4) = 4 \end{cases}$$

Put $m = 2,\ n = 3$

$$f(6) = f(2) \,.\, f(3) \begin{cases} \text{when } f(2) = 1 \\ f(3) = 1 \text{ to } 7 \\ f(2) = 2 \\ f(3) = 1 \text{ or } 2 \text{ or } 3 \end{cases}$$

$f(5), f(7)$ can take any value
Total $= (1 \times 1 \times 7 \times 1 \times 7 \times 1 \times 7) + (1 \times 1 \times 3 \times 1 \times 7 \times 1 \times 7)$
$N = 490$
Sum of digits of N = 13

Q85: Let $f : [0, 3] \rightarrow R$ be define by

$f\left(x\right) = \min\left\{x - [x], 1 + [x] - x\right\}$

where $[x]$is the greatest integer less than or equal to x. Let P denote the set containing all $x \in [0, 3]$ where f is discontinuous, and Q denote the set containing all $x \in (0, 3)$where f is not differentiable. Then the sum of number of elements in P and Q is equal to __________.

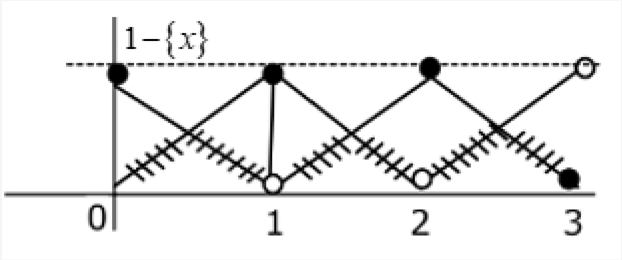
Q85: Let $f:[0,3] \rightarrow R$ be define by

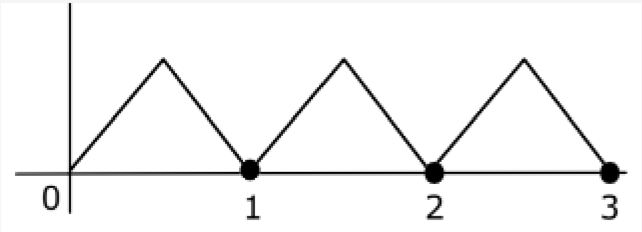
$f(x)=\min \{x-[x], 1+[x]-x\}$

where $[x]$is the greatest integer less than or equal to x. Let P denote the set containing all $x \in [0,3]$ where f is discontinuous, and Q denote the set containing all $x \in (0,3)$where f is not differentiable. Then the sum of number of elements in P and Q is equal to ________.

5.00

Solution:





Non differentiable at

$x=\frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}$ and continuous $\forall\ x \in [0,3]$

Q86: For real numbers α and β, consider the following system of linear equations: $x + y - z = 2,\ x + 2y + \alpha z = 1, 2x - y + z = \beta$. If the system has infinite solutions, then $\alpha + \beta$ is equal to ______.

Q86: For real numbers α and β, consider the following system of linear equations: $x+y-z=2,\ x+2y+\alpha z=1, 2x-y+z=\beta$. If the system has infinite solutions, then $\alpha+\beta$ is equal to ______.

5.00

Solution:

For infinite solutions

$\Delta=\Delta_1=\Delta_2=\Delta_3=0$

$$\Delta=\begin{vmatrix}1 & 1 & -1\\ 1 & 2 & \alpha\\ 2 & -1 & 1\end{vmatrix}=0$$

$R_1\rightarrow R_1+R_3$

$$\Rightarrow\Delta=\begin{vmatrix}3 & 0 & 0\\ 1 & 2 & \alpha\\ 2 & -1 & 1\end{vmatrix}=0$$

$\Rightarrow\Delta=3(2+\alpha)=0$

$\Rightarrow\alpha=-2$

$$\Delta_2=\begin{vmatrix}1 & 2 & -1\\ 1 & 1 & -2\\ 2 & \beta & 1\end{vmatrix}=0$$

$1(1+2\beta)-2(1+4)-(\beta-2)=0$

$\beta-7=0$

$\beta=7$

$\therefore\alpha+\beta=5$ Ans.

Q87: Let the domain of the function $f(x) = \log_4(\log_5(\log_3(18x - x^2 - 77)))$ be (a, b). Then the value of the integral $\int_a^b \frac{\sin^3 x}{(\sin^3 x + \sin^3(a+b-x))} dx$ is equal to.

Q87: Let the domain of the function $f(x) = \log_4\left(\log_5\left(\log_3\left(18x - x^2 - 77\right)\right)\right)$ be (a, b). Then the value of the integral $\int\limits_a^b \frac{\sin^3 x}{\left(\sin^3 x + \sin^3 (a+b-x)\right)} dx$ is equal to.

1.00

Solution:

For domain

$\log_5\left(\log_3\left(18x - x^2 - 77\right)\right) > 0$

$\log_3\left(18x - x^2 - 77\right) > 1$

$18x - x^2 - 77 > 3$

$x^2 - 18x + 80 < 0$

$x \in (8, 10)$

$\Rightarrow a = 8$ and $b = 10$

$I = \int\limits_a^b \frac{\sin^3 x}{\sin^3 x + \sin^3 (a+b-x)} dx$

$I = \int\limits_a^b \frac{\sin^3 (a+b-x)}{\sin^3 x + \sin^3 (a+b-x)}$

$2I = (b - a) \Rightarrow I = \frac{b-a}{2} (\because\ a\ = 8 \text{ and b } =\ 10)$

$I = \frac{10-8}{2} = 1$

Q88: If $\log_3 2,\ \log_3\left(2^x-5\right), \log_3\left(2^x-\frac{7}{2}\right)$ are in an arithmetic progression, then the value of x is equal to______.

Q88: If $\log_3 2,\ \log_3 (2^x - 5), \log_3 \left(2^x - \frac{7}{2}\right)$ are in an arithmetic progression, then the value of x is equal to_____.

3.00

Solution:

$2\log_3 (2^x - 5) = \log_3 2 \left(2^x - \frac{7}{2}\right)$

Let $2^x = t$

$\log_3 (t-5)^2 = \log_3 2 \left(t - \frac{7}{2}\right)$

$(t-5)^2 = 2t - 7$

$t^2 - 12t + 32 = 0$

$(t-4)(t-8) = 0$

$\Rightarrow 2^x = 4$ or $2^x = 8$

$x = 2$ (Rejected)

or $x = 3$

Q89: If $y = y\left(x\right), y \in \left[0, \frac{\pi}{2}\right)$ is the solution of the differential equation $\frac{dy}{dx} - \sin(x + y) - \sin(x - y) = 0$, with $y\left(0\right) = 0$, then $5y'\left(\frac{\pi}{2}\right)$ is equal to _____.

Q89: If $y = y(x), y \in \left[0, \frac{\pi}{2}\right)$ is the solution of the differential equation $\frac{dy}{dx} - \sin(x+y) - \sin(x-y) = 0$, with $y(0) = 0$, then $5y'\left(\frac{\pi}{2}\right)$ is equal to ______.

2.00

Solution:

$\sec y \frac{dy}{dx} = 2 \sin x \cos y$
$\sec^2 y dy = 2 \sin x dx$
$\tan y = -2 \cos x + c$
$y(0) = 0$ gives $c = 2$
$\tan y = -2 \cos x + 2 \Rightarrow$ at $x = \frac{\pi}{2}$
$\tan y = 2$
$\sec^2 y \frac{dy}{dx} = 2 \sin x$
$5 \frac{dy}{dx} = 2$

Q90: Let $\overrightarrow{a} = \hat{i} + \hat{j} + \hat{k}, \hat{b}$ and $\overrightarrow{c} = \hat{j} - \hat{k}$ be three vectors such that $\overrightarrow{a} \times \overrightarrow{b} = \overrightarrow{c}$ and $\overrightarrow{a} . \overrightarrow{b} = 1$. If the length of projection vector of the vector $\overrightarrow{b}$ on the vector $\overrightarrow{a} \times \overrightarrow{c}$ is l, then the value of $3l^2$ is equal to ______.

Q90: Let $\overrightarrow{a} = \hat{i} + \hat{j} + \hat{k}, \hat{b}$ and $\overrightarrow{c} = \hat{j} - \hat{k}$ be three vectors such that $\overrightarrow{a} \times \overrightarrow{b} = \overrightarrow{c}$ and $\overrightarrow{a} . \overrightarrow{b} = 1$. If the length of projection vector of the vector $\overrightarrow{b}$ on the vector $\overrightarrow{a} \times \overrightarrow{c}$ is l, then the value of $3l^2$ is equal to _______.

2.00

Solution:

$\vec{a} \times \vec{b} = c$

Take Dot with $\overrightarrow{c}$

$(a \times b) \cdot c = c \cdot c$

$[a \quad b \quad c] = |c|^2 = 2$

Projection of $\overrightarrow{b}$ on $\overrightarrow{a} \times \overrightarrow{c} = \ell$

$$\frac{\left|\overrightarrow{b}.\left(\overrightarrow{a} \times \overrightarrow{c}\right)\right|}{\left|\overrightarrow{a} \times \overrightarrow{c}\right|} = \ell$$

$\therefore \ell = \frac{2}{\sqrt{6}} \Rightarrow \ell^2 = \frac{4}{6}$

$3\ell^2 = 2$